EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Li-6800 STOMATAL CONDUCTANCE MODEL FITTING

- This script will attempt to fit the methods of Woning et al (2026) and McAusland et al (2016) using the script supplied by Silvere Vialet-Chabrand
- The models fitted in the script look to be the Cumulative Distribution Weibull (CDWeibull) models described in Eqn 4.1 of Woning et al (2026)
- I've also adapted the script to incorporate the McAusland model
- Scripts supplied by Silvere can be found in the gs_fit.zip directory
- There are two scripts in this directory: fit_per_curve.r and fit_per_species.r
- Fit_per_curve looks to take each individual plant and fit a curve to it, while fit_per_species fits some parameters (tau and theta) at a species (treatment in our setting) level and others at a replicate level
- In an email from Silvere (shared by Chris and put in the *Materials from Ed* directory), Silvere prefers the fit_per_species approach, as do I, as this seems to align better with conventional analysis approaches

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.li6800.stomatalCondModel.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/")
library(dae)
library(dplyr)
library(growthPheno) # calcLagged
packageVersion("growthPheno")
```

```
[1] '3.1.18'
```

```
library(brms) ##package used to fit bayesian regression model using Stan
```

Set up constants and files

```
options(width=90)
load("../data/li6800.v1.Rdata")
saveFile <- "../data/stomatalCondModel.v1.Rdata"
```

Define TW to be unique for each treatment

```
li6800.dat$TW <- paste(li6800.dat$Temperature, li6800.dat$Watering, sep="_")
li6800.dat$TW <- factor(li6800.dat$TW)
```

Define th and tmin, to be time in hours and minutes respectively

```
li6800.dat$th <- li6800.dat$xElapsed/3600
li6800.dat$tmin <- li6800.dat$xElapsed/60
```

Explore the gs profiles of just the light on portion of the experiment

```
plotProfiles(li6800.dat %>% filter(Light == "On"), response = "gsw",
             individuals = "TWR", times = "tmin",
```

```
              facet.x = "Temperature",
              facet.y = "Watering",
              y.title = "gsw",
              x.title = "hours",
              title = "gsw",
              breaks.spacing.x = 5,
              alpha = 0.75)
```

Explore the gs profiles of just the second light off portion of the experiment

```
plotProfiles(li6800.dat %>% filter(Light == "Off" & th > 2), response = "gsw",
              individuals = "TWR", times = "th",
              facet.x = "Temperature",
              facet.y = "Watering",
              y.title = "gsw",
              x.title = "hours",
              title = "gsw",
              breaks.spacing.x = 5,
              alpha = 0.75)
```

Following *fit_per_species.R*

- Script begins by specifying a stomatal conductance model and setting priors on the parameters
- Am using the same form of the model and priors as those specified in the script - no changes made
- Following discussion with Ed on 22/04, perhaps we want to select a different "training" dataset for each treatment combination - need to explore what impact this has

```
nlform <- bf(gsw ~ g1 + (g0 - g1) * exp(-((th / tau)^theta)),
              g0 ~ 1 + (1 | Rep),
              g1 ~ 1 + (1 | Rep),
              tau ~ 1,
              theta ~ 1,
              nl = TRUE)
```

Stomatal conductance model - CDWeibull from Woning et al 2026

```
nlform_mc <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k)+1)),
              g0 ~ 1 + (1 | Rep),
              g1 ~ 1 + (1 | Rep),
              k ~ 1,
              lambda ~ 1,
              nl = TRUE)

priors <- c(prior(normal(0.025, 0.02), nlpar = "g0"),
             prior(normal(0.3, 0.2), nlpar = "g1"),
             prior(normal(0, 0.01), class = "sd", nlpar = "g0"),
             prior(normal(0, 0.05), class = "sd", nlpar = "g1"),
             prior(normal(0.25, 1), nlpar = "tau", lb = 0),
             prior(normal(2, 2), nlpar = "theta", lb = 0),
             prior(student_t(3, 0, 0.01), class = "sigma"))
```

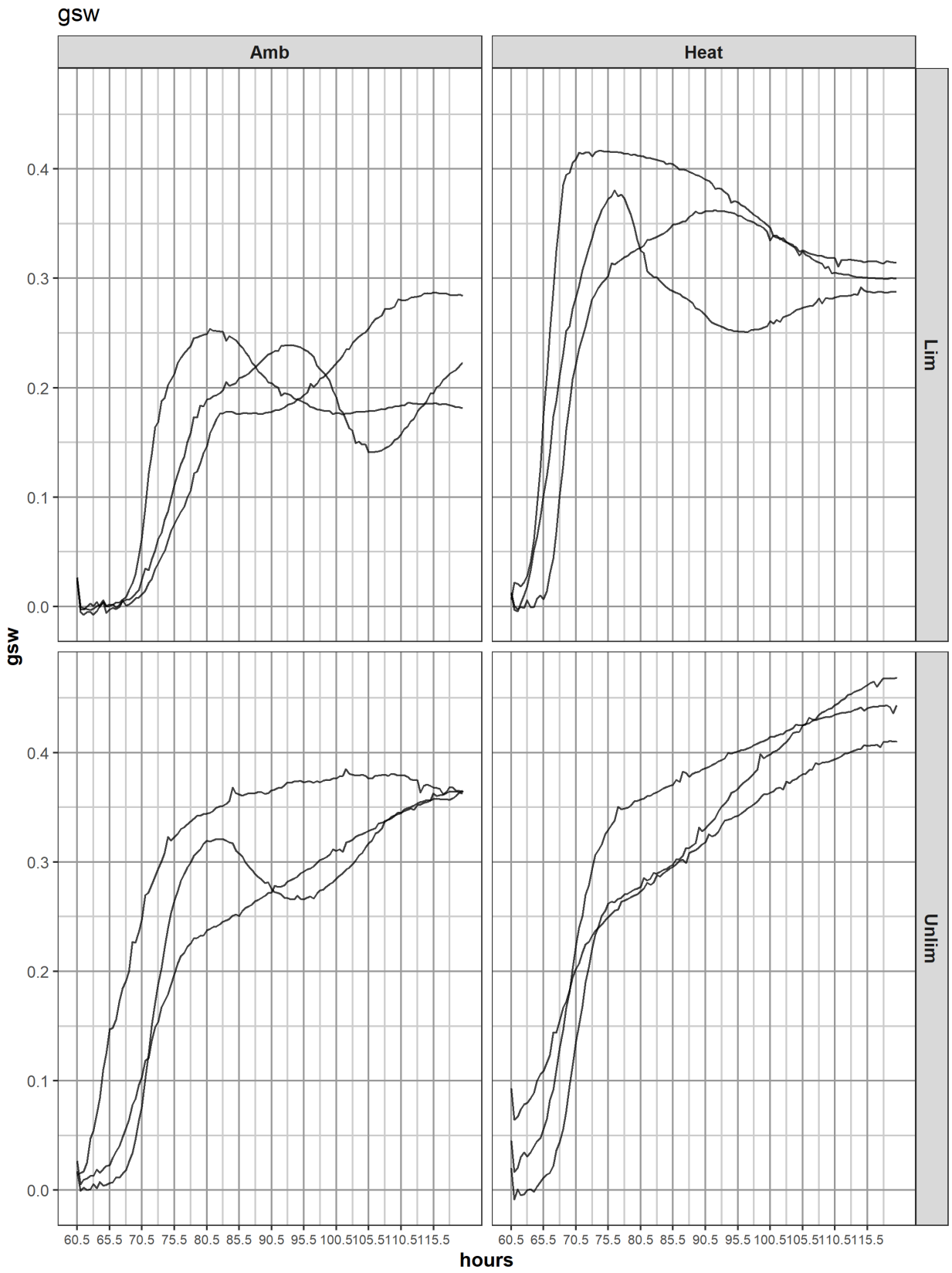


Figure 1: plot of chunk unnamed-chunk-5

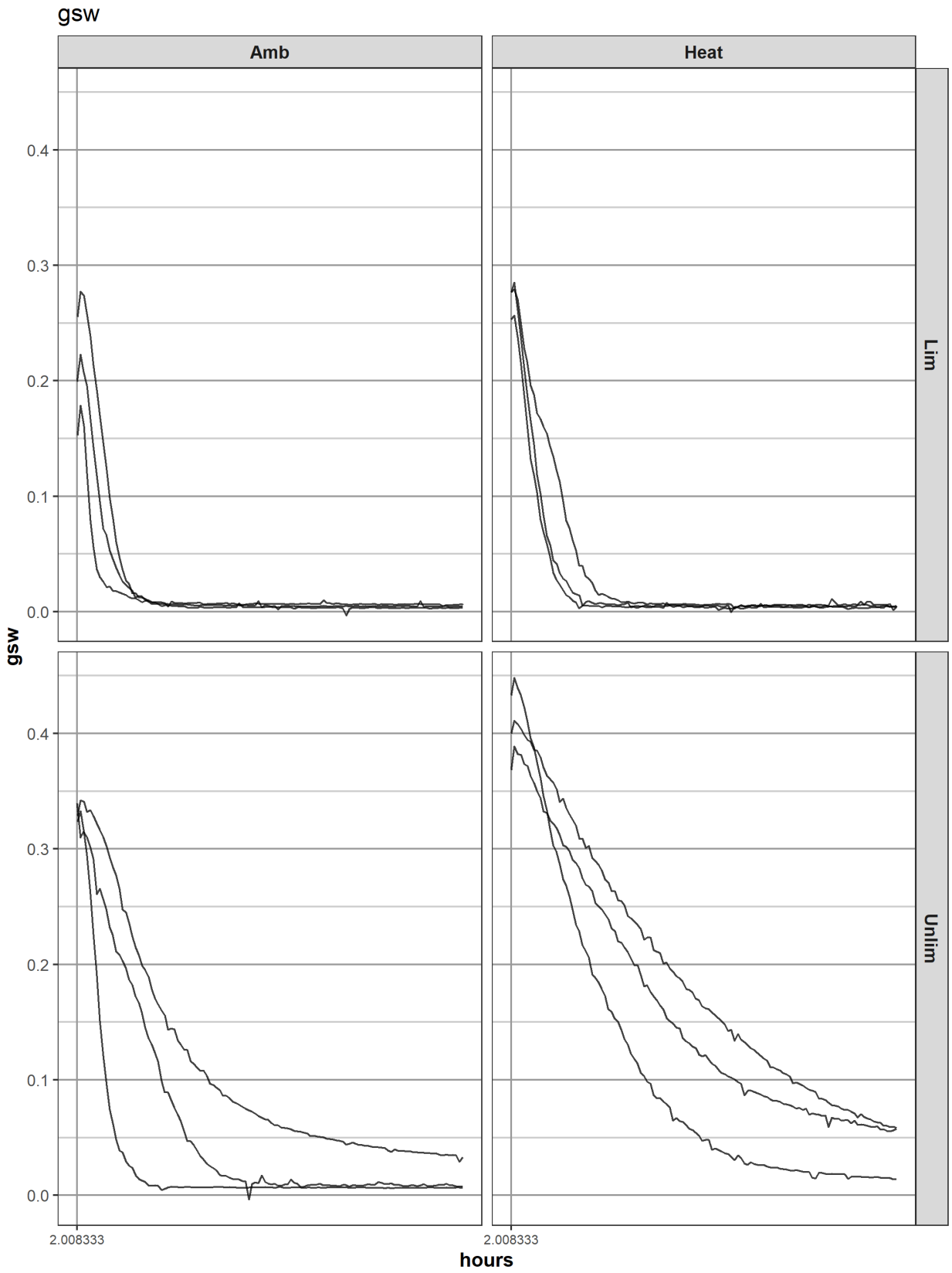


Figure 2: plot of chunk unnamed-chunk-6

```
priors_mc <- c(prior(normal(0.025, 0.02), nlpar = "g0"),
               prior(normal(0.3, 0.2), nlpar = "g1"),
               prior(normal(0, 0.01), class = "sd", nlpar = "g0"),
               prior(normal(0, 0.05), class = "sd", nlpar = "g1"),
               prior(normal(3, 2), nlpar = "k", lb = 0),
               prior(normal(4, 2), nlpar = "lambda", lb = 0),
               prior(student_t(3, 0, 0.01), class = "sigma"))
```

Stomatal conductance model - McAusland from McAusland et al 2016

Initialise the model Looks like to initialise the model, the dataframe is subset for the first plant in the first rep and for PAR == 1, which is the period where the light is switched on

```
# pick a small but valid dataset
dd_init <- subset(li6800.dat, Light == "On" & TW == li6800.dat$TW[1] & th != 0)
dd_init$th <- dd_init$th - dd_init$th[1]
dd_init$tmin <- dd_init$tmin - dd_init$tmin[1]

##remove the first observation from the analysis - it's still impacted by the change in lighting
dd_init <- subset(dd_init, th != 0)
head(dd_init)
```

```
    obs         time elapsed             date   hhmmss       time.elapsed        A
125 126 1750032134.5    3661 20250616 10:02:14 10:02:14 61.000000000000021 1.296567
126 127 1750032164.5    3691 20250616 10:02:44 10:02:44 61.500000000000021 1.385702
127 128 1750032194.5    3721 20250616 10:03:14 10:03:14 62.000000000000021 1.067099
128 129 1750032224.5    3751 20250616 10:03:44 10:03:44 62.500000000000014 1.284658
129 130 1750032254.5    3781 20250616 10:04:14 10:04:14 63.000000000000014 1.230181
130 131 1750032284.5    3811 20250616 10:04:44 10:04:44 63.500000000000014 1.372948
              gsw  TleafEB     light           Timestamp time.diff xElapsed xElapsed.min
125 -0.0028748183 24.75610  999.9787 2025-06-16 10:02:14        30     3660         61.0
126 -0.0025839934 24.84859  999.9951 2025-06-16 10:02:44        30     3690         61.5
127 -0.0001403046 24.91740 1000.0007 2025-06-16 10:03:14        30     3720         62.0
128  0.0026056428 24.98566 1000.0335 2025-06-16 10:03:44        30     3750         62.5
129  0.0005737584 25.07066 1000.0050 2025-06-16 10:04:14        30     3780         63.0
130  0.0044335864 25.10999 1000.0035 2025-06-16 10:04:44        30     3810         63.5
    Line Temperature Watering Rep       TWR Location Light      TW          th tmin
125  L01         Amb      Lim   1 Amb_Lim_1      D13    On Amb_Lim 0.008333333  0.5
126  L01         Amb      Lim   1 Amb_Lim_1      D13    On Amb_Lim 0.016666667  1.0
127  L01         Amb      Lim   1 Amb_Lim_1      D13    On Amb_Lim 0.025000000  1.5
128  L01         Amb      Lim   1 Amb_Lim_1      D13    On Amb_Lim 0.033333333  2.0
129  L01         Amb      Lim   1 Amb_Lim_1      D13    On Amb_Lim 0.041666667  2.5
130  L01         Amb      Lim   1 Amb_Lim_1      D13    On Amb_Lim 0.050000000  3.0
```

```
plotProfiles(dd_init, response = "gsw",
             individuals = "TWR", times = "tmin",
             y.title = "gsw",
             x.title = "hours",
             title = "gsw",
             breaks.spacing.x = 5,
             alpha = 0.75)
```

```
# Configure and compile the model
fit_init <- brm(
  formula = nlform,
```

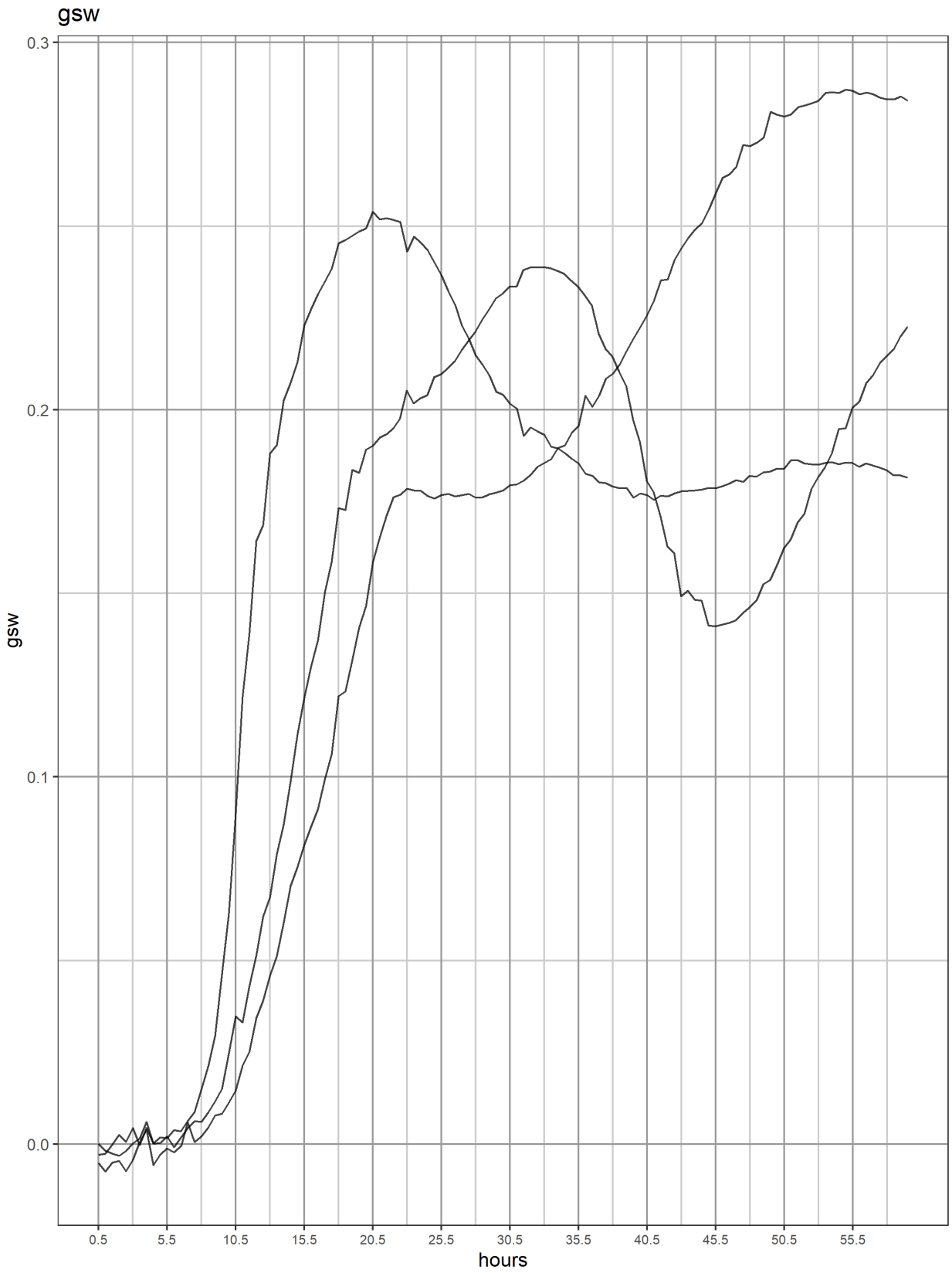


Figure 3: plot of chunk unnamed-chunk-9

```
  data = dd_init,
  prior = priors,
  chains = 4,
  cores = 4,
  iter = 3000,
  warmup = 2000,
  control = list(
    adapt_delta = 0.99,
    max_treedepth = 12,
    metric = "diag_e"
  )
)
```

```
Compiling Stan program...
```

```
Start sampling
```

```
Warning: There were 2 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.
```

```
Warning: Examine the pairs() plot to diagnose sampling problems
```

```
print(summary(fit_init))
```

```
Warning: There were 2 divergent transitions after warmup. Increasing adapt_delta above 0.99 may help. Se
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g1 + (g0 - g1) * exp(-((th/tau)^theta))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         tau ~ 1
         theta ~ 1
   Data: dd_init (Number of observations: 362)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.01     0.01     0.03 1.00     2309     2583
sd(g1_Intercept)     0.02      0.02     0.01     0.07 1.00     1145     1524

Regression Coefficients:
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept       0.00      0.01    -0.01     0.03 1.00     1969     2385
g1_Intercept       0.21      0.02     0.17     0.25 1.00     1084     1151
tau_Intercept      0.26      0.01     0.24     0.27 1.00     3610     2622
theta_Intercept    3.63      0.39     2.94     4.46 1.00     3990     2812

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.03     0.04 1.01     4037     2783

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
```

```
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(fit_init)
```

```
pairs(fit_init)
```

```
mcmc_plot(fit_init, type = "trace")
```

```
rstan::check_hmc_diagnostics(fit_init$fit)
```

```
Divergences:
2 of 4000 iterations ended with a divergence (0.05%).
Try increasing 'adapt_delta' to remove the divergences.

Tree depth:
0 of 4000 iterations saturated the maximum tree depth of 12.

Energy:
E-BFMI indicated no pathological behavior.
```

```
loo(fit_init)
```

```
Computed from 4000 by 362 log-likelihood matrix.

         Estimate   SE
elpd_loo    682.6 11.4
p_loo         8.0  0.6
looic     -1365.2 22.8
------
MCSE of elpd_loo is 0.0.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.2]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```
pp_check(fit_init)  # compares density of the observed data with data generated from posterior predicti
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fit_init, type = "error_hist", ndraws = 12)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
pp_check(fit_init, type = "scatter_avg", ndraws = 1000)
```

```
pp_check(fit_init, type = "stat_2d")
```

```
Using all posterior draws for ppc type 'stat_2d' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.
```

```
pp_check(fit_init, type = "loo_ribbon")
```

```
Using all posterior draws for ppc type 'loo_ribbon' by default.
```

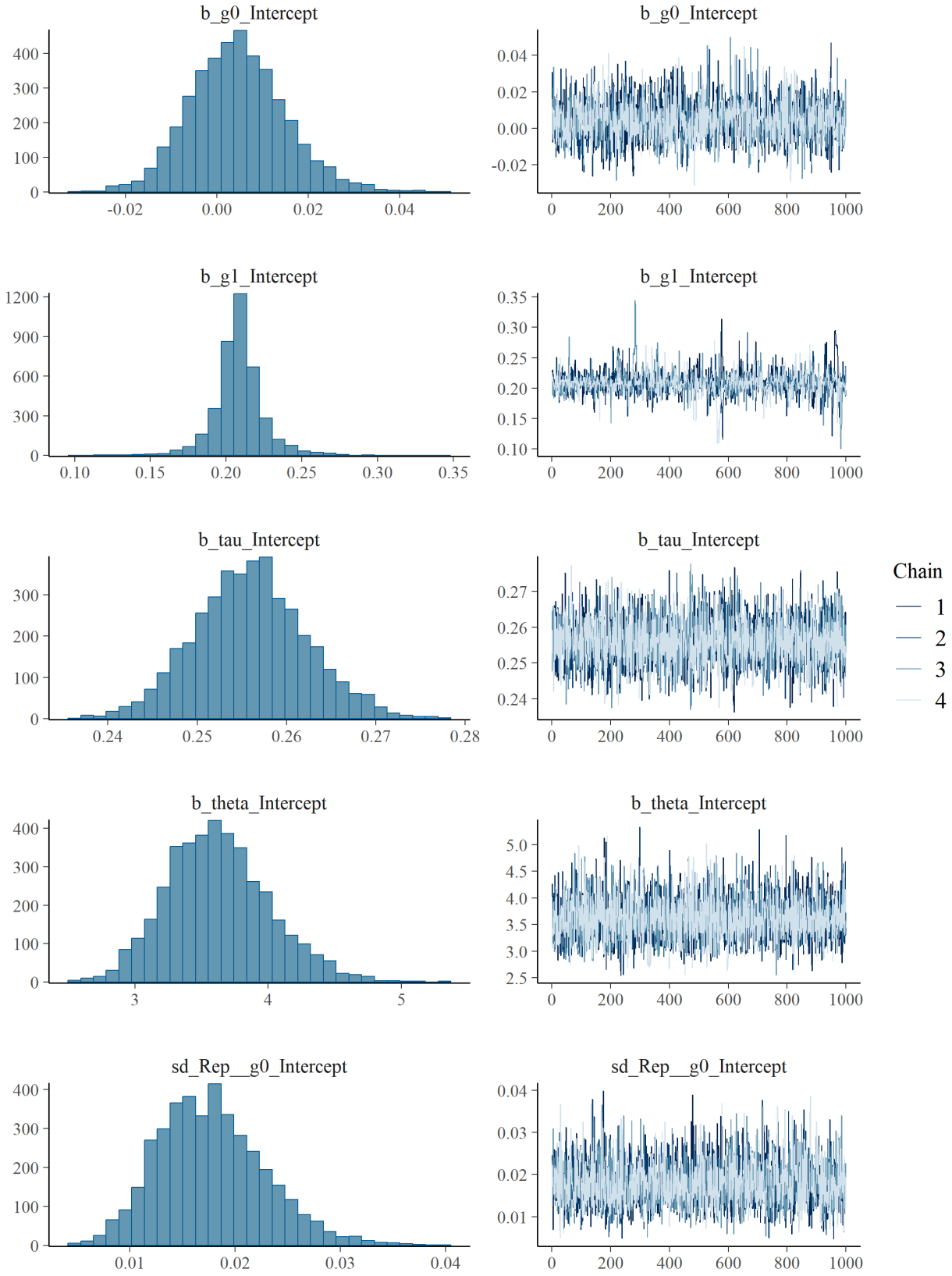


Figure 4: plot of chunk unnamed-chunk-9

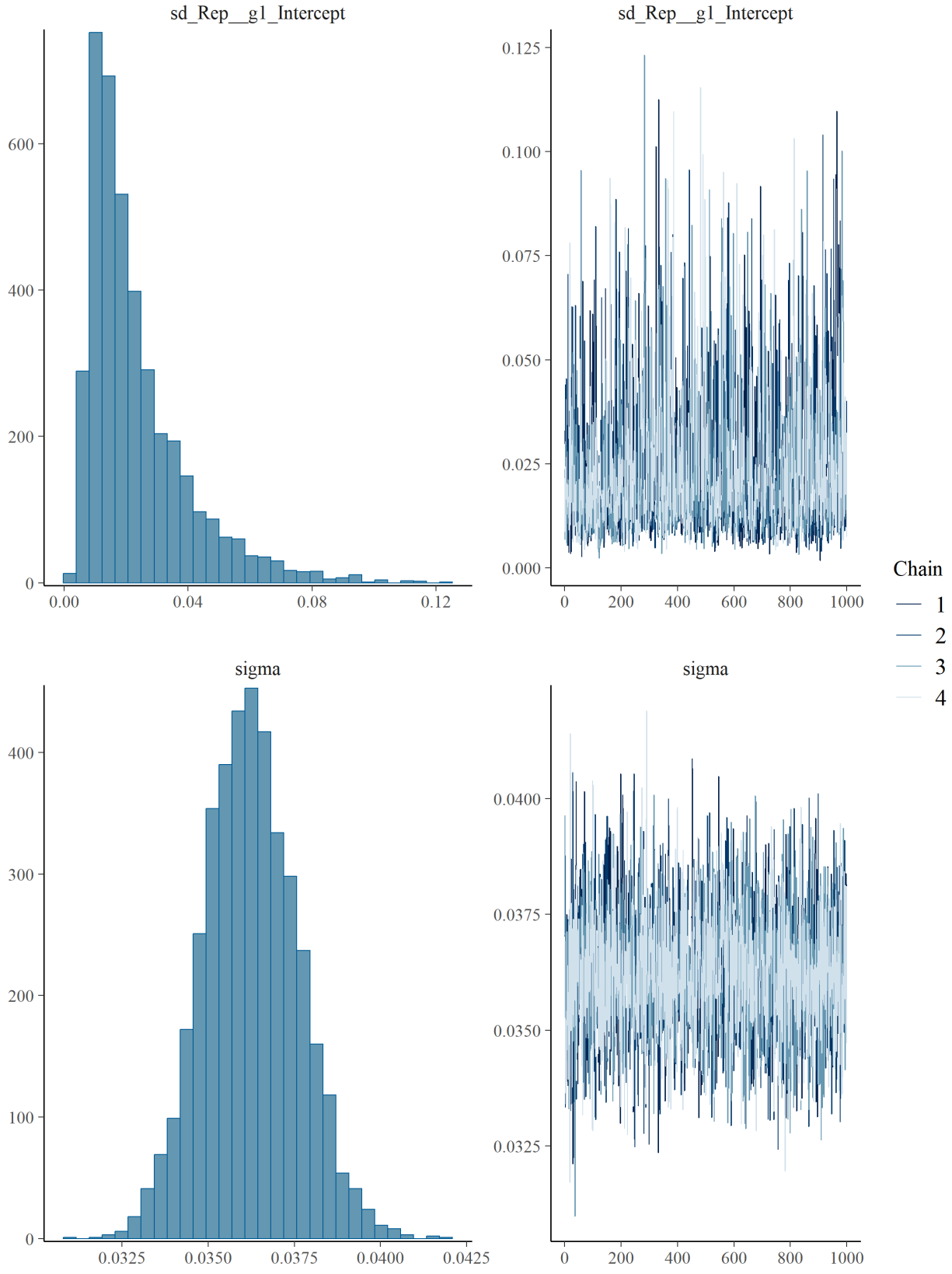


Figure 5: plot of chunk unnamed-chunk-9

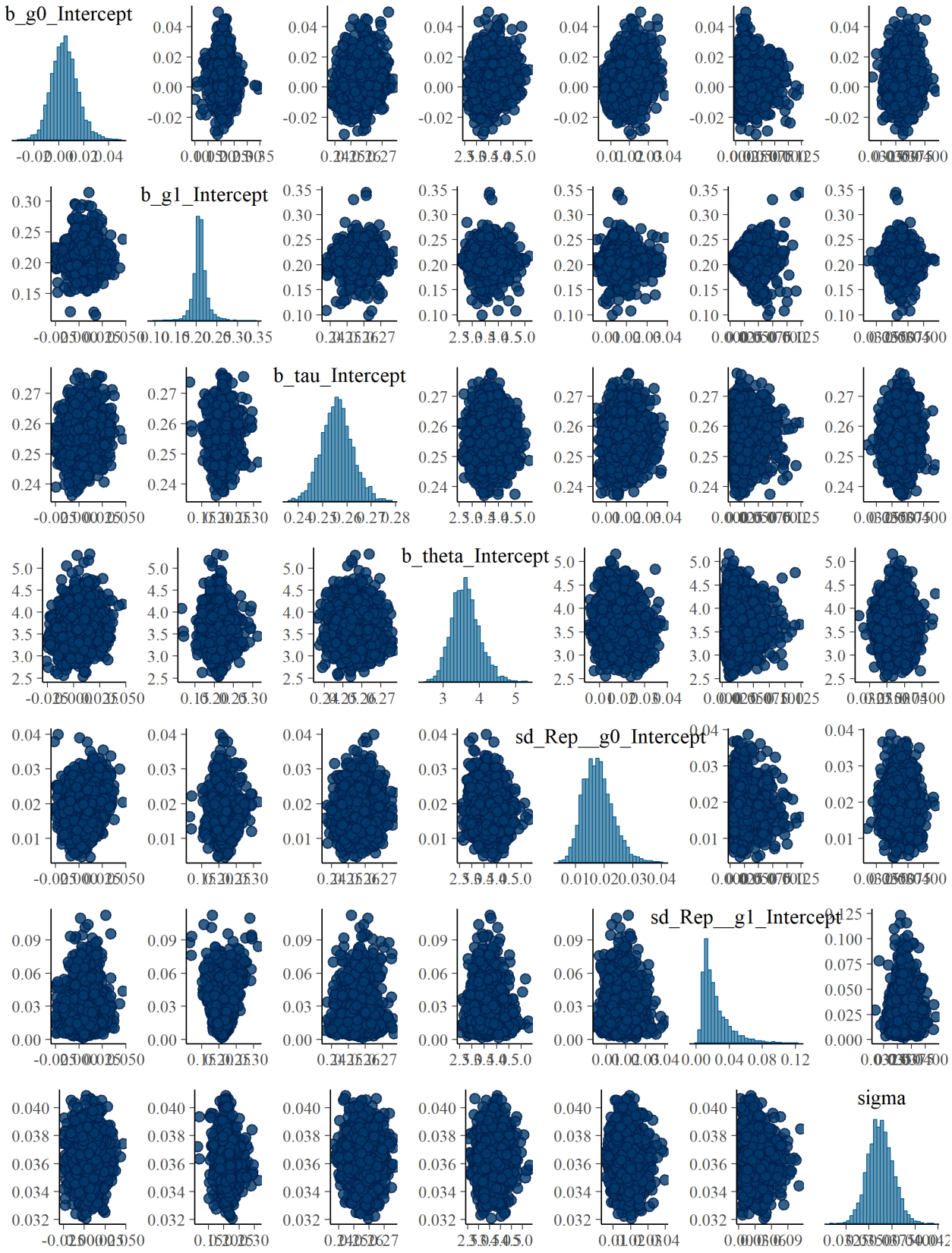


Figure 6: plot of chunk unnamed-chunk-9

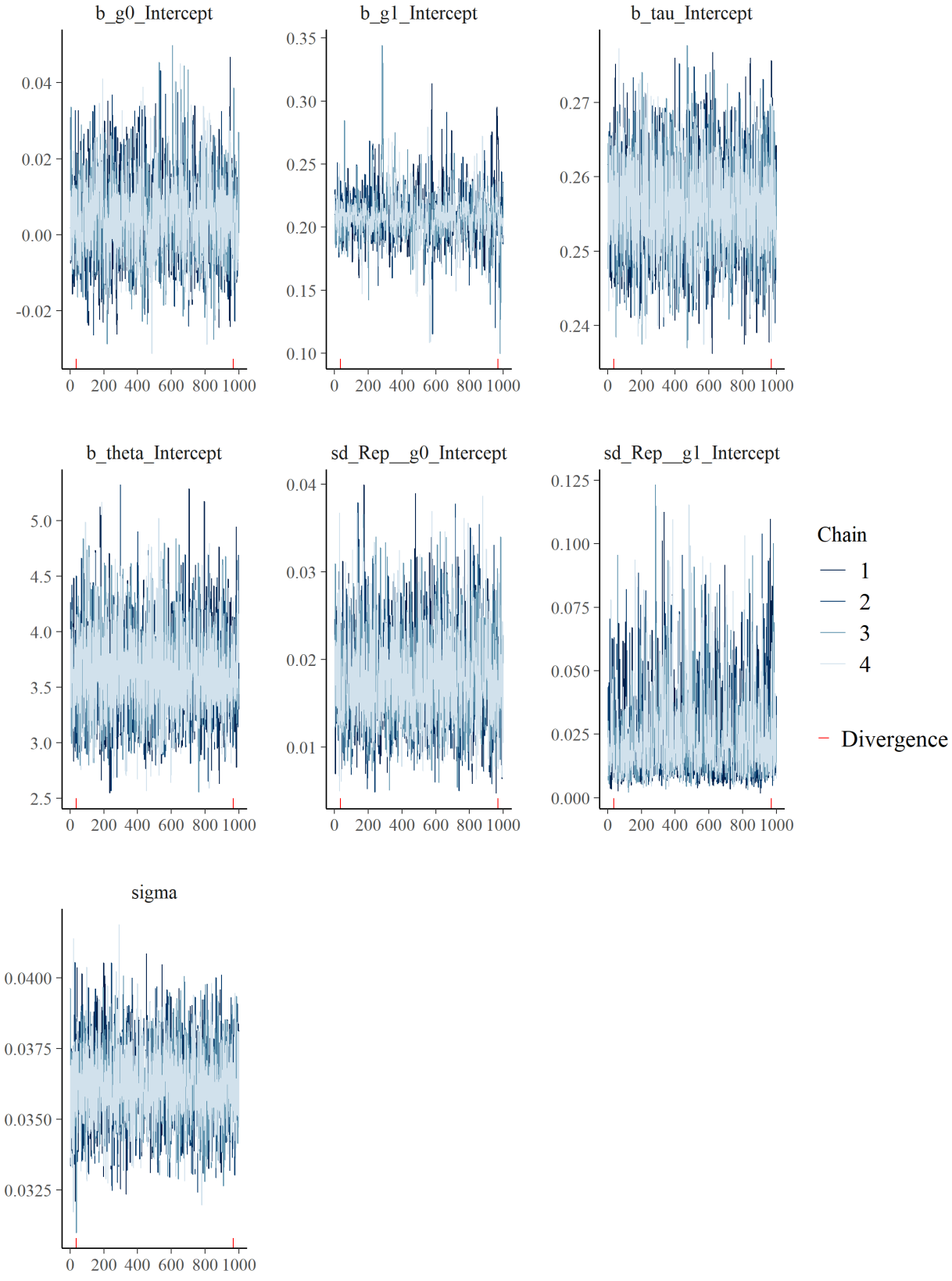


Figure 7: plot of chunk unnamed-chunk-9

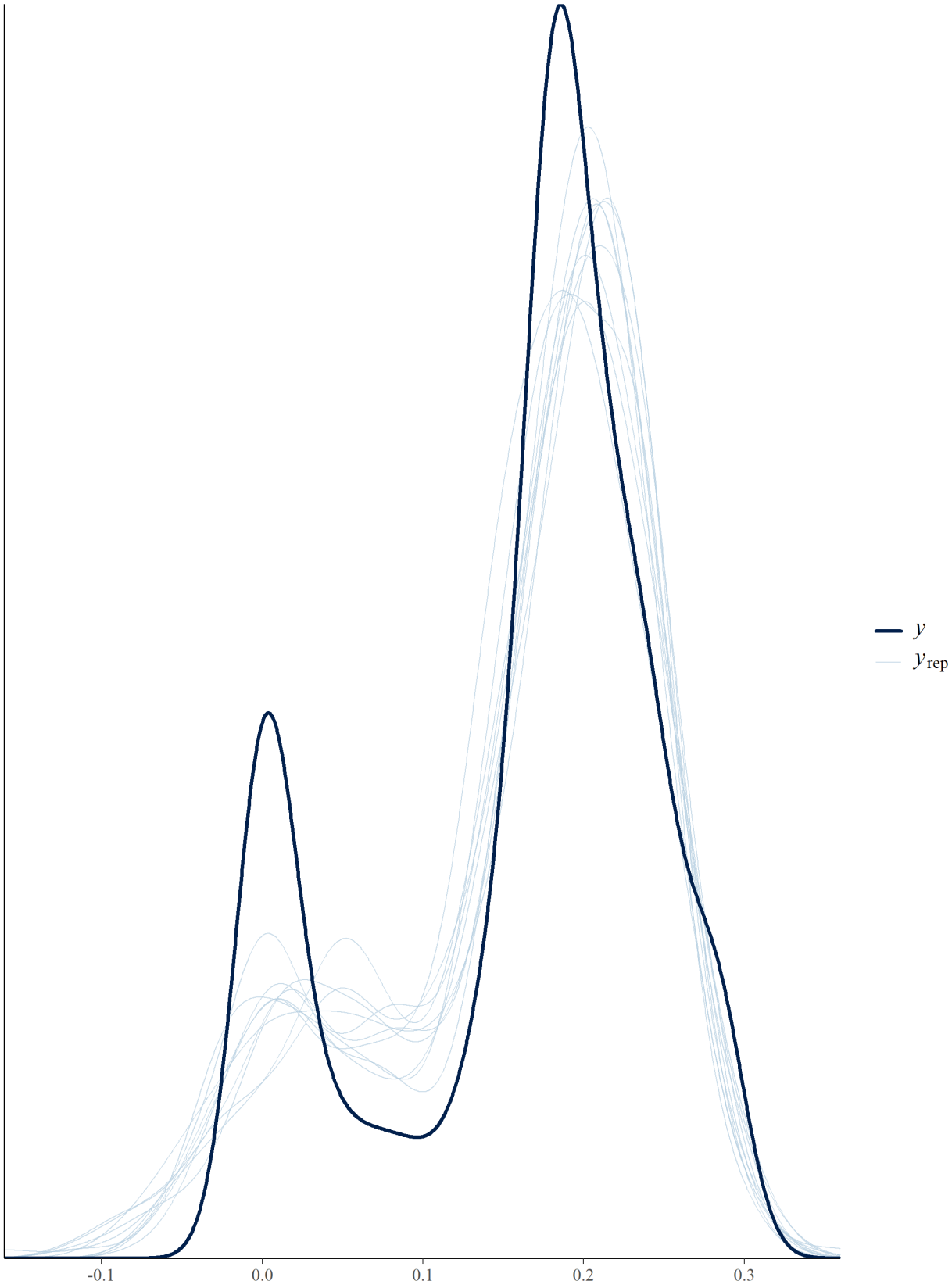


Figure 8: plot of chunk unnamed-chunk-9

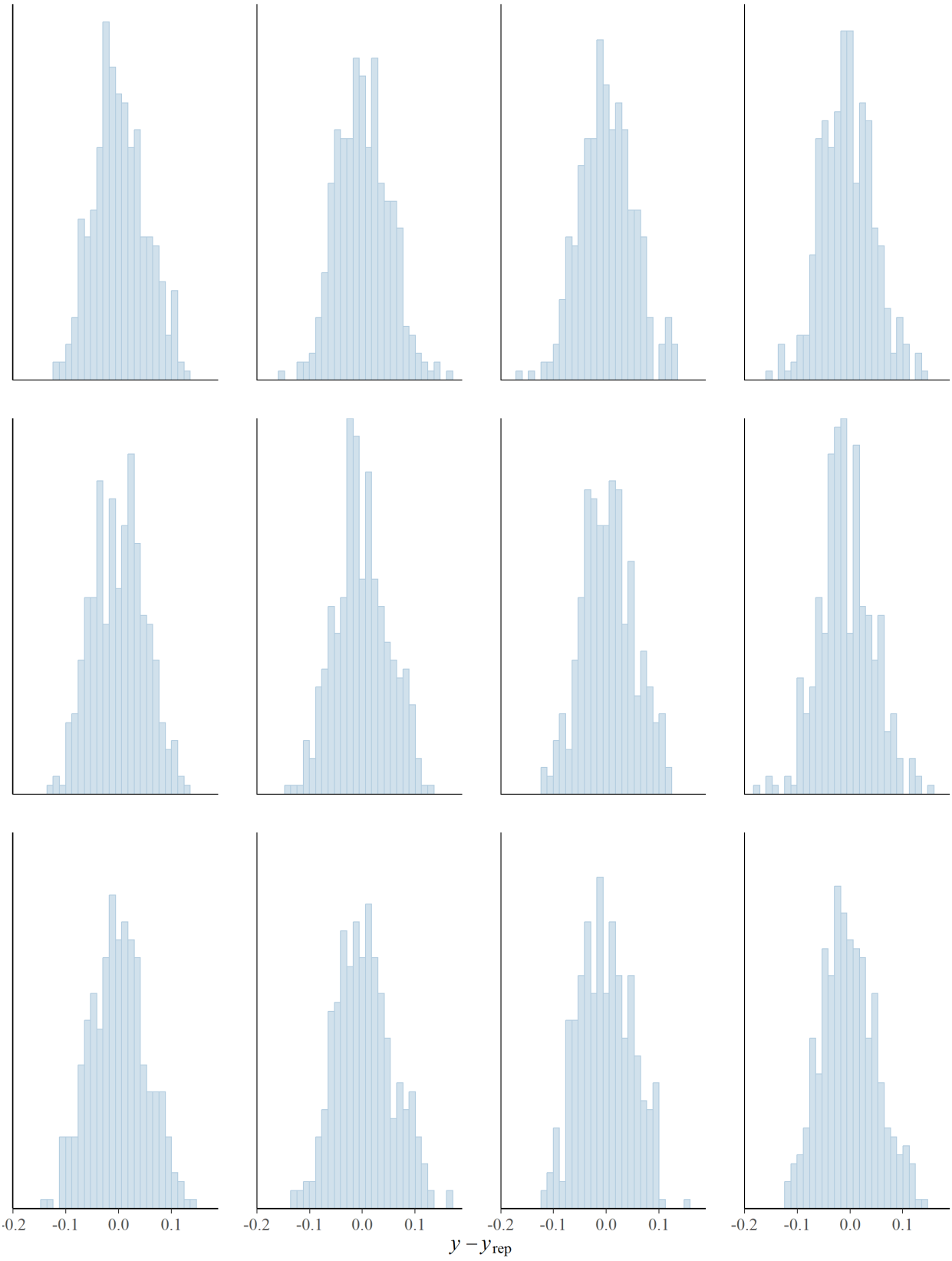


Figure 9: plot of chunk unnamed-chunk-9

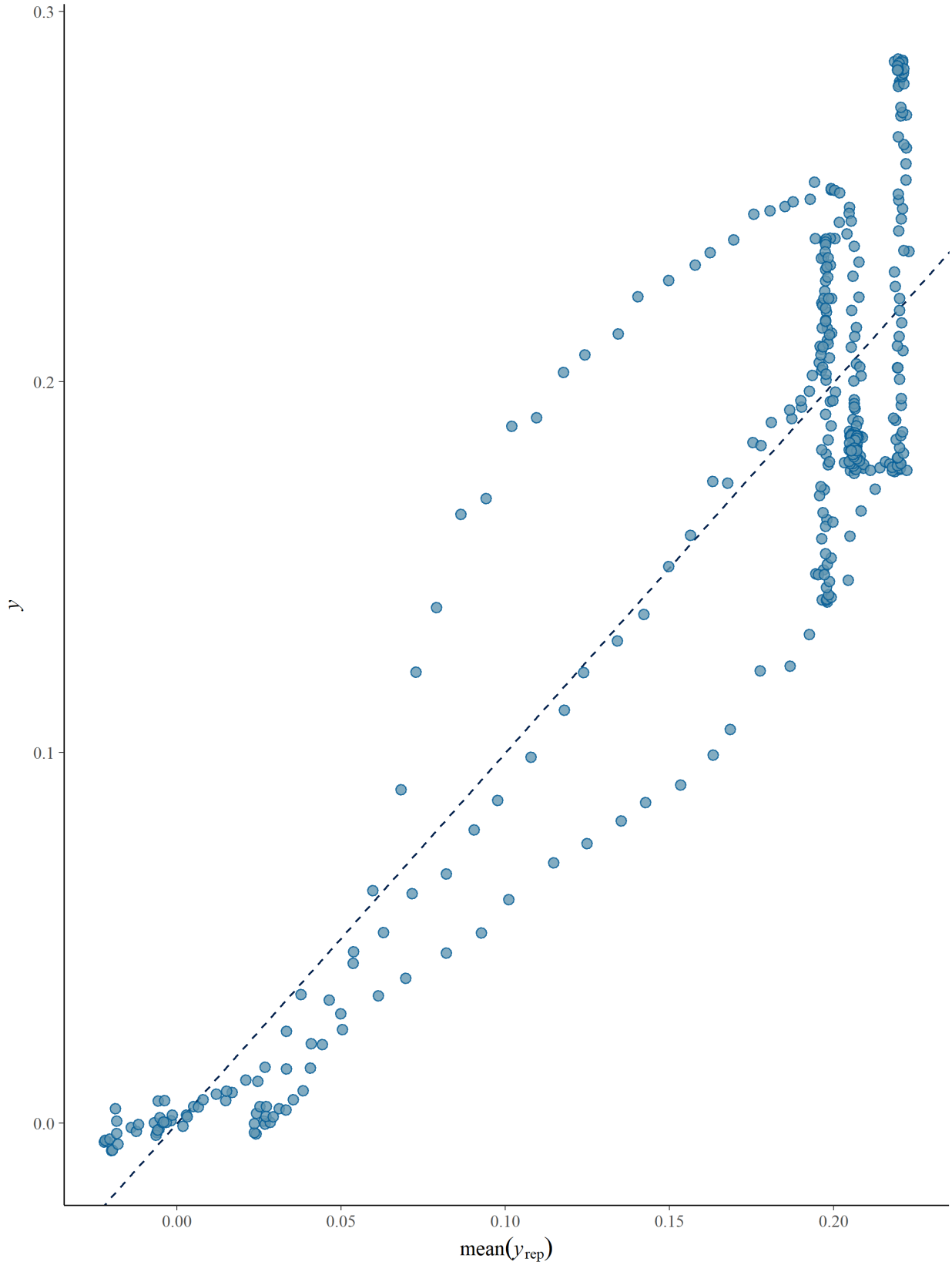


Figure 10: plot of chunk unnamed-chunk-9

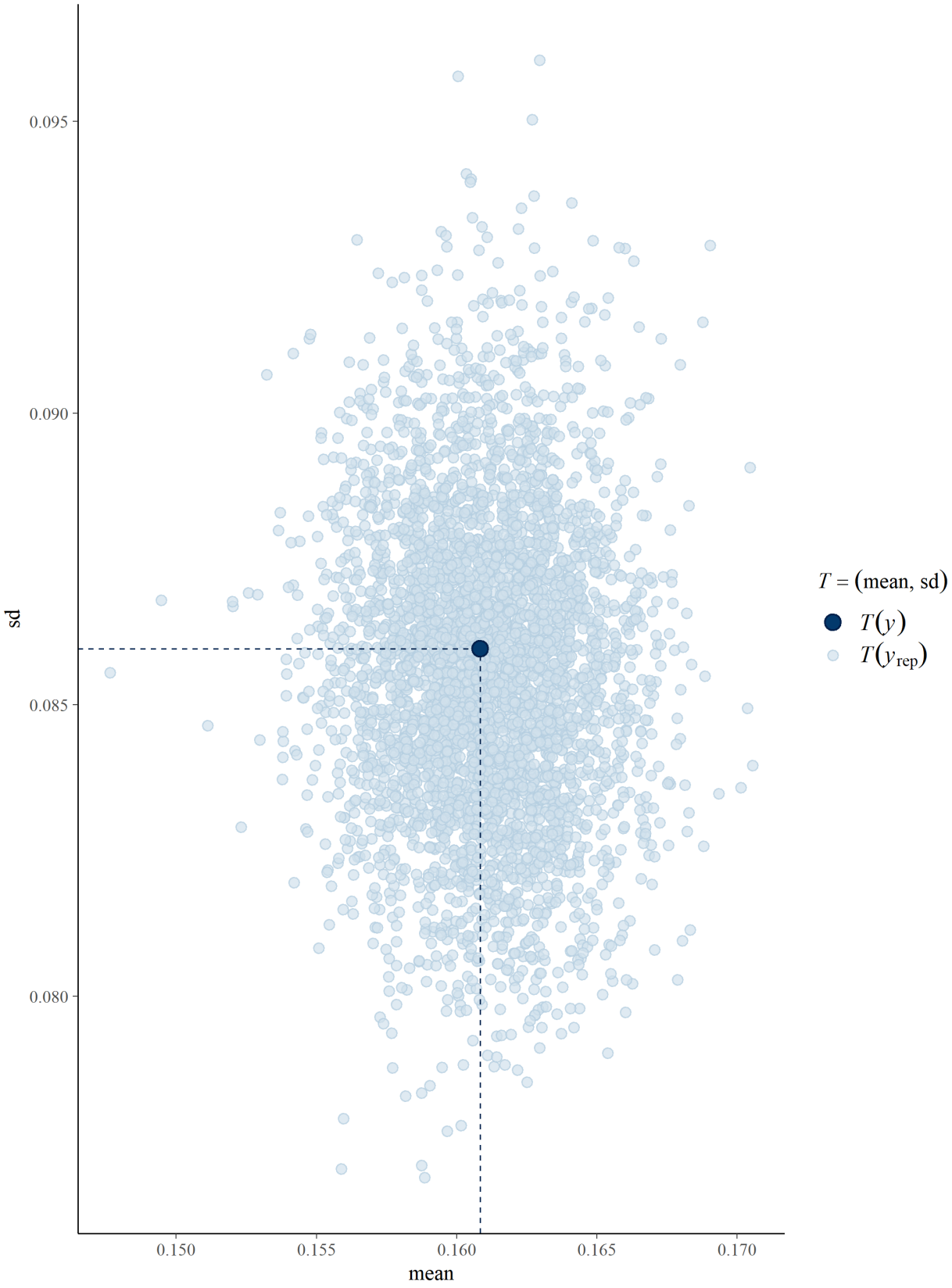


Figure 11: plot of chunk unnamed-chunk-9

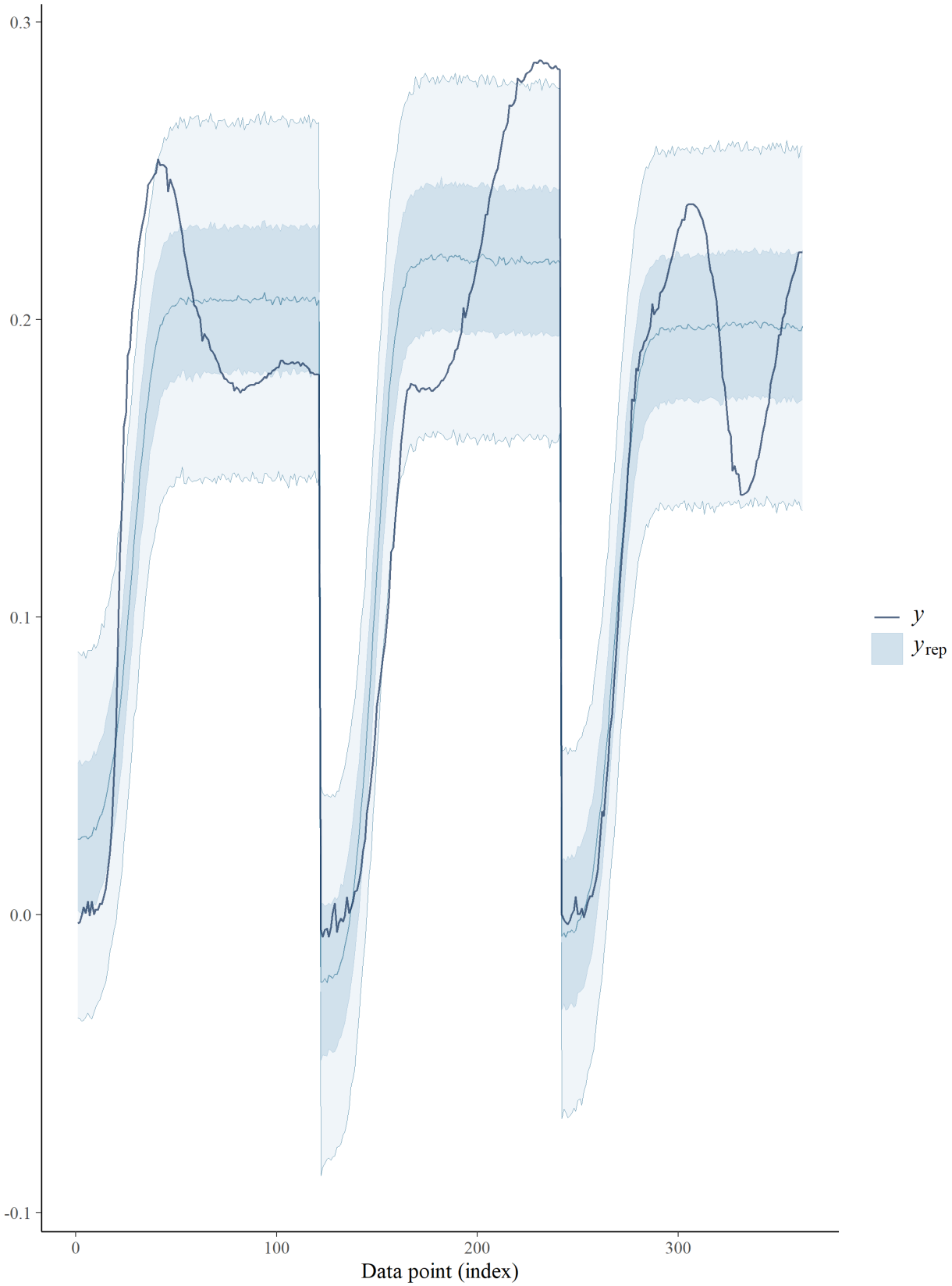


Figure 12: plot of chunk unnamed-chunk-9

```r
plot(gsw~th, dd_init, pch=16, main = unique(dd_init$TW))
points(fitted(fit_init)[,1]~th, dd_init, pch=16, col="red")
```

```r
# Configure and compile the model - McAusland
fit_init_mc <- brm(
  formula = nlform_mc,
  data = dd_init,
  prior = priors_mc,
  chains = 4,
  cores = 4,
  iter = 3000,
  warmup = 2000,
  control = list(
    adapt_delta = 0.99,
    max_treedepth = 12,
    metric = "diag_e"
  )
)
```

```
Compiling Stan program...
```

```
Start sampling
```

```r
print(summary(fit_init_mc))
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         k ~ 1
         lambda ~ 1
   Data: dd_init (Number of observations: 362)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.00     0.01     0.03 1.00     2188     2709
sd(g1_Intercept)     0.02      0.02     0.01     0.07 1.00     1357     1998

Regression Coefficients:
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept         0.01      0.01    -0.01     0.03 1.00     1966     2294
g1_Intercept         0.21      0.02     0.18     0.25 1.00     1262     1427
k_Intercept          3.88      0.38     3.17     4.67 1.00     2834     3015
lambda_Intercept     8.25      0.59     7.03     9.38 1.00     2753     2692

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.03     0.04 1.00     3395     2256

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
```

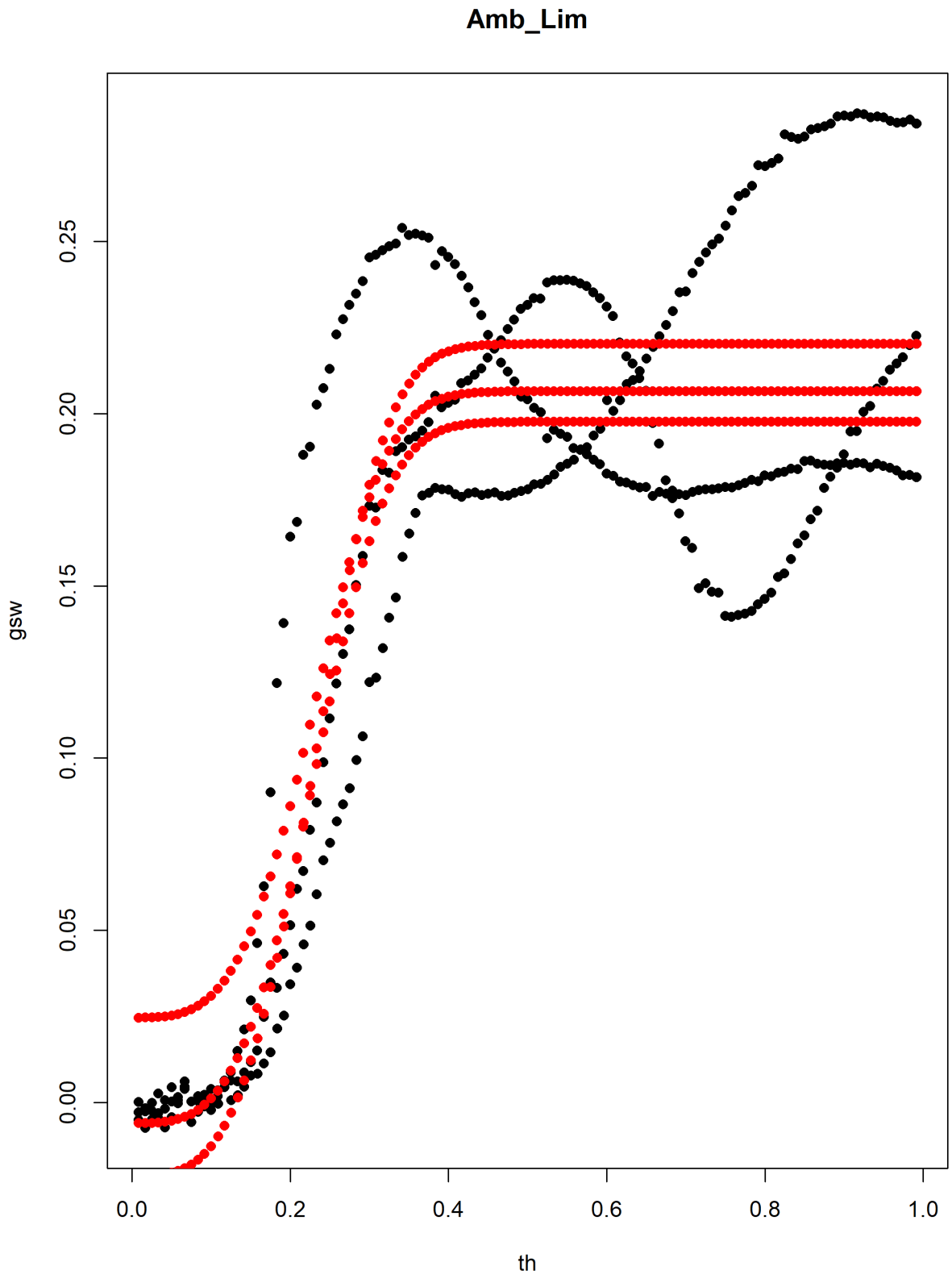


Figure 13: plot of chunk unnamed-chunk-9

```
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(fit_init_mc)
```

```
pairs(fit_init_mc)
```

```
mcmc_plot(fit_init_mc, type = "trace")
```

```
No divergences to plot.
```

```
rstan::check_hmc_diagnostics(fit_init_mc$fit)
```

```
Divergences:
0 of 4000 iterations ended with a divergence.

Tree depth:
0 of 4000 iterations saturated the maximum tree depth of 12.

Energy:
E-BFMI indicated no pathological behavior.
```

```
loo(fit_init_mc)
```

```
Computed from 4000 by 362 log-likelihood matrix.

         Estimate   SE
elpd_loo    683.7 10.7
p_loo         7.5  0.5
looic     -1367.3 21.5
------
MCSE of elpd_loo is 0.0.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.3]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```
pp_check(fit_init_mc)  # compares density of the observed data with data generated from posterior predi
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fit_init_mc, type = "error_hist", ndraws = 12)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
pp_check(fit_init_mc, type = "scatter_avg", ndraws = 1000)
```

```
pp_check(fit_init_mc, type = "stat_2d")
```

```
Using all posterior draws for ppc type 'stat_2d' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.
```

```
pp_check(fit_init_mc, type = "loo_ribbon")
```

```
Using all posterior draws for ppc type 'loo_ribbon' by default.
```

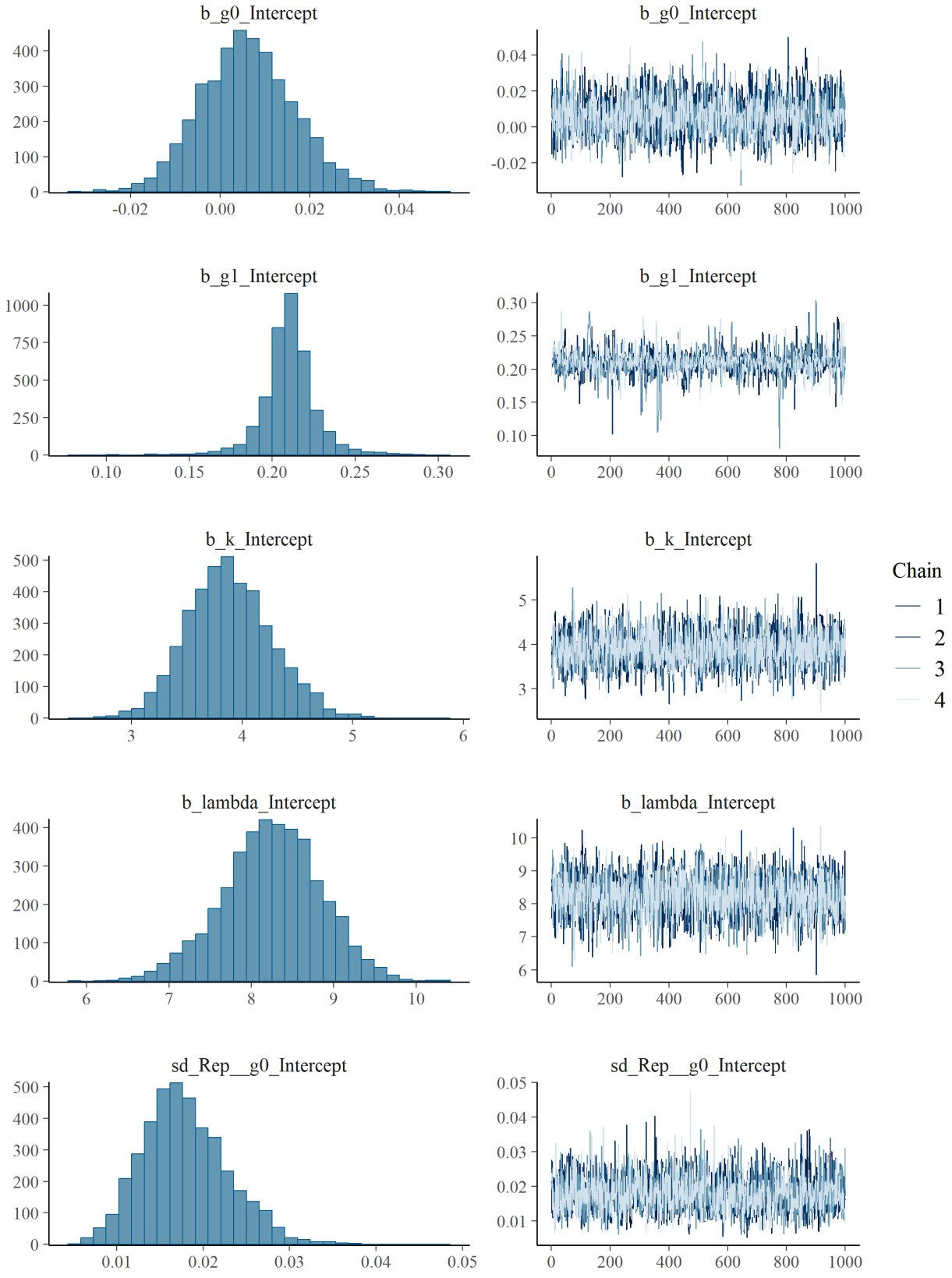


Figure 14: plot of chunk unnamed-chunk-9

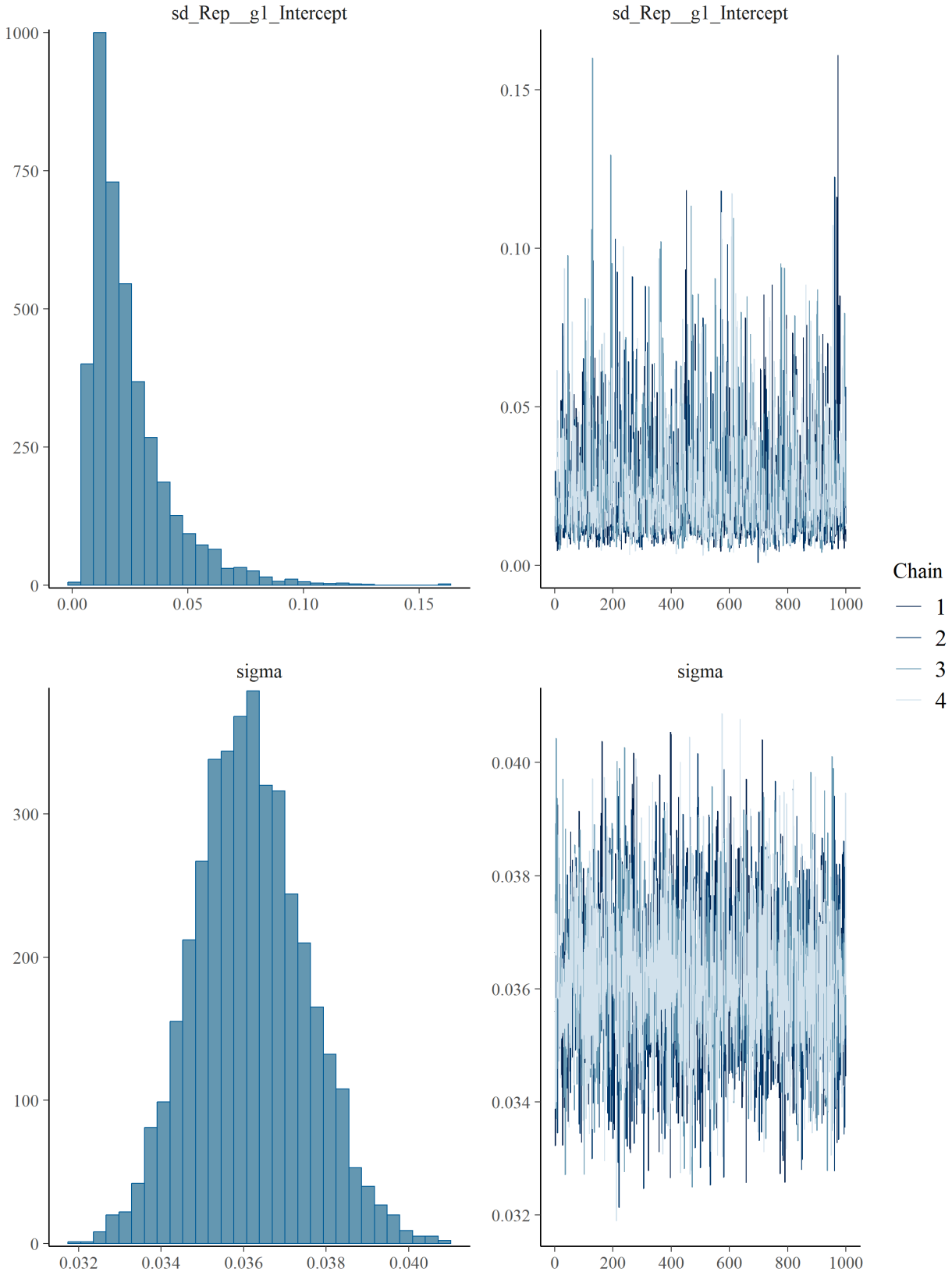


Figure 15: plot of chunk unnamed-chunk-9

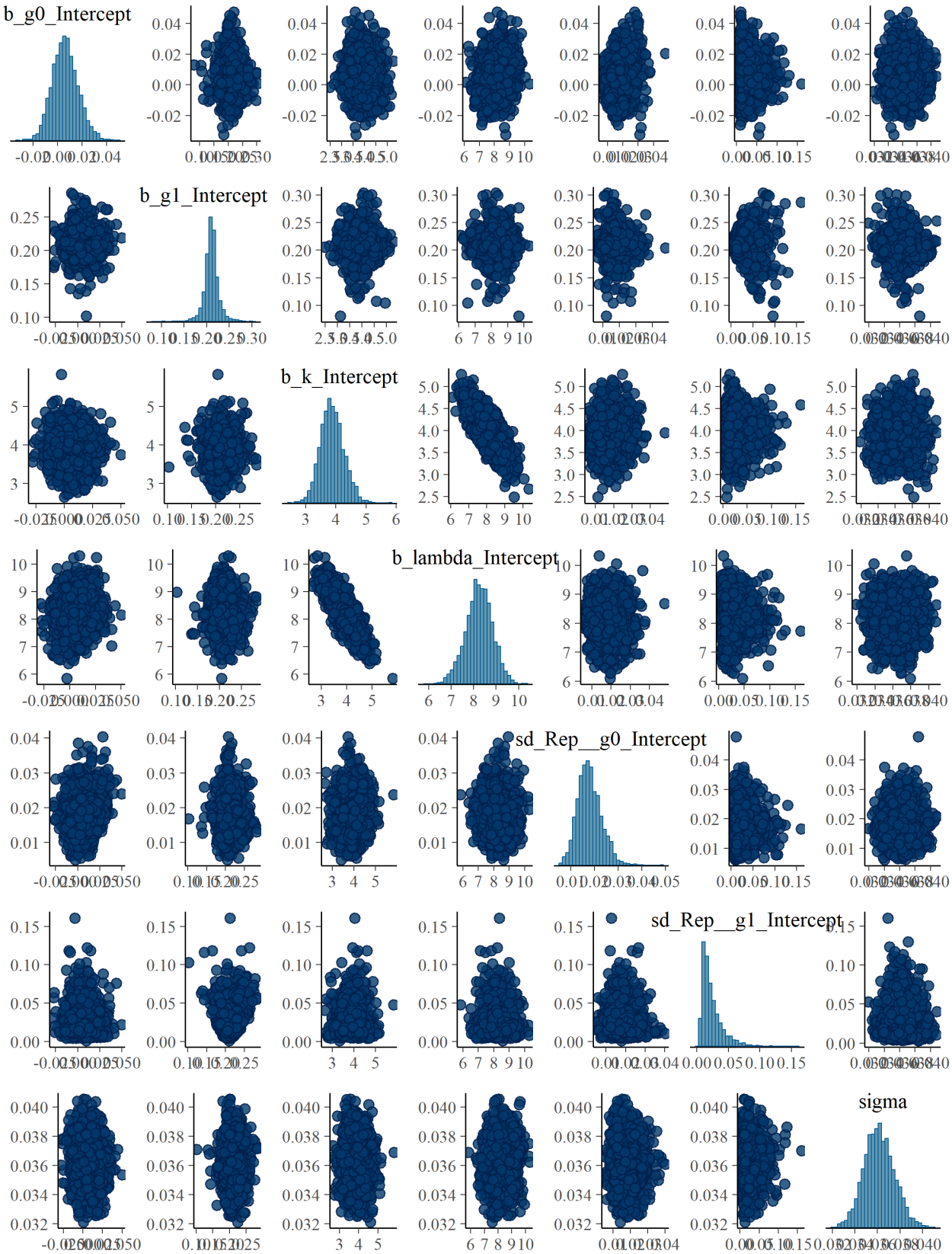


Figure 16: plot of chunk unnamed-chunk-9

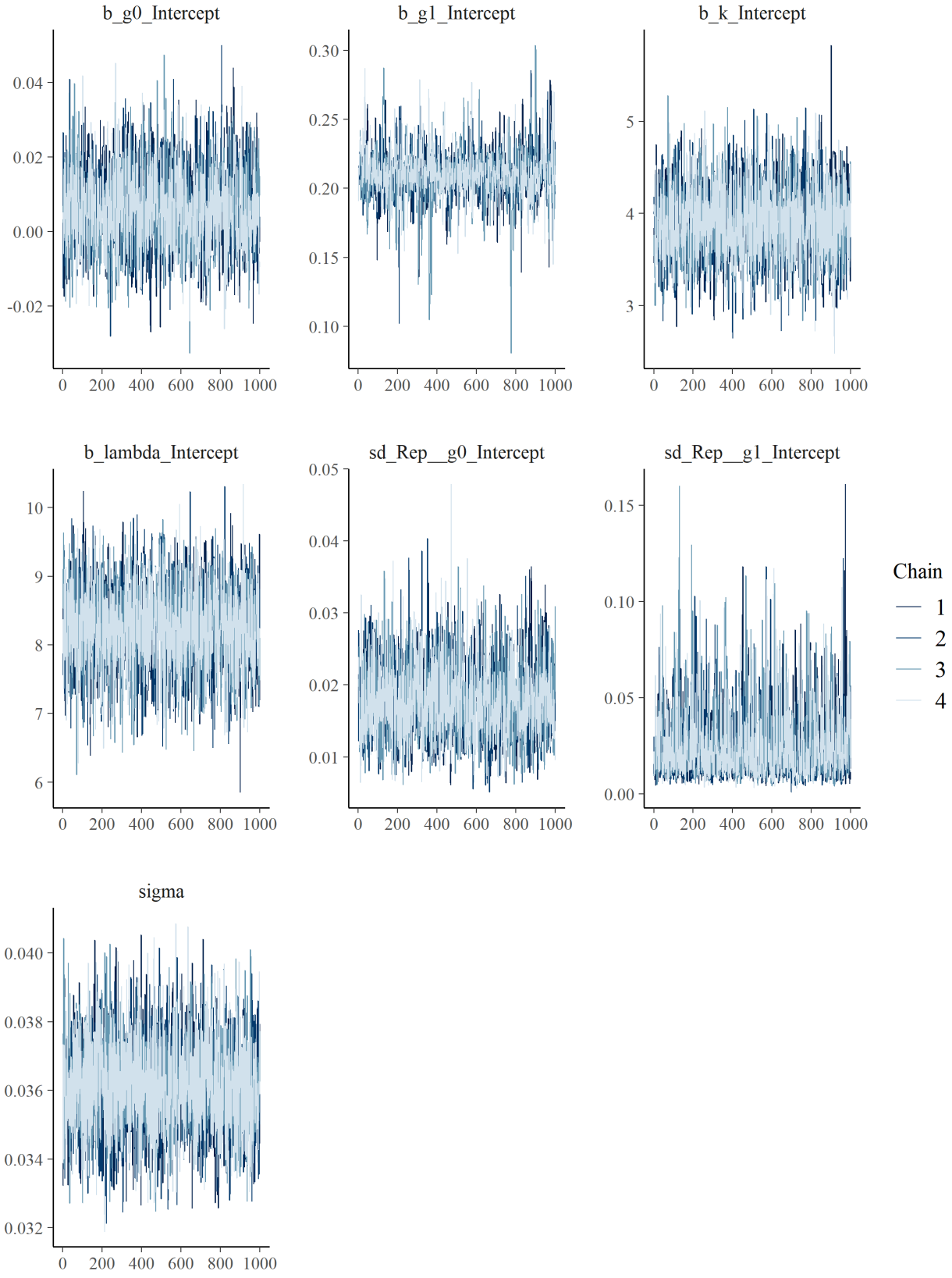


Figure 17: plot of chunk unnamed-chunk-9

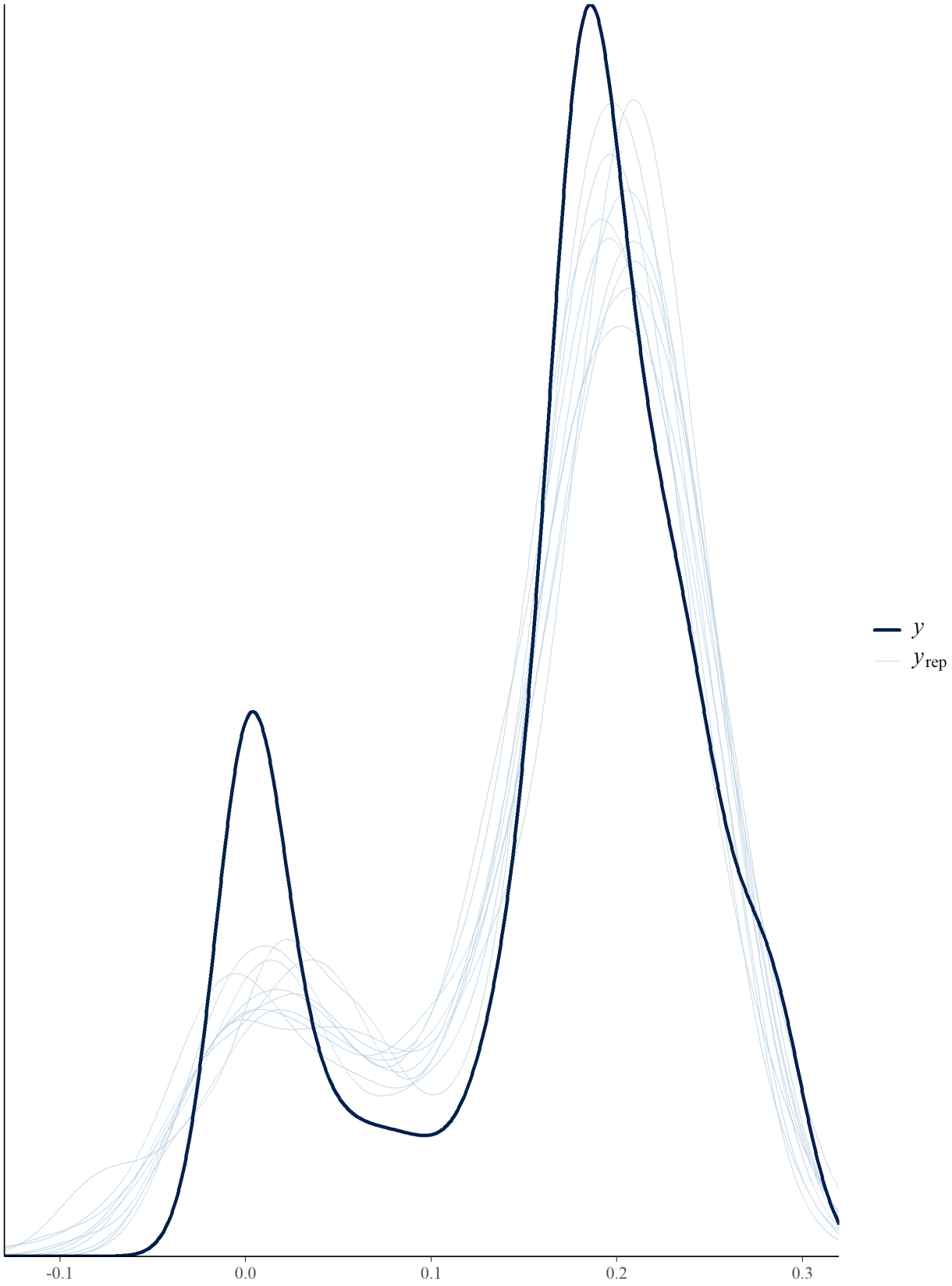


Figure 18: plot of chunk unnamed-chunk-9

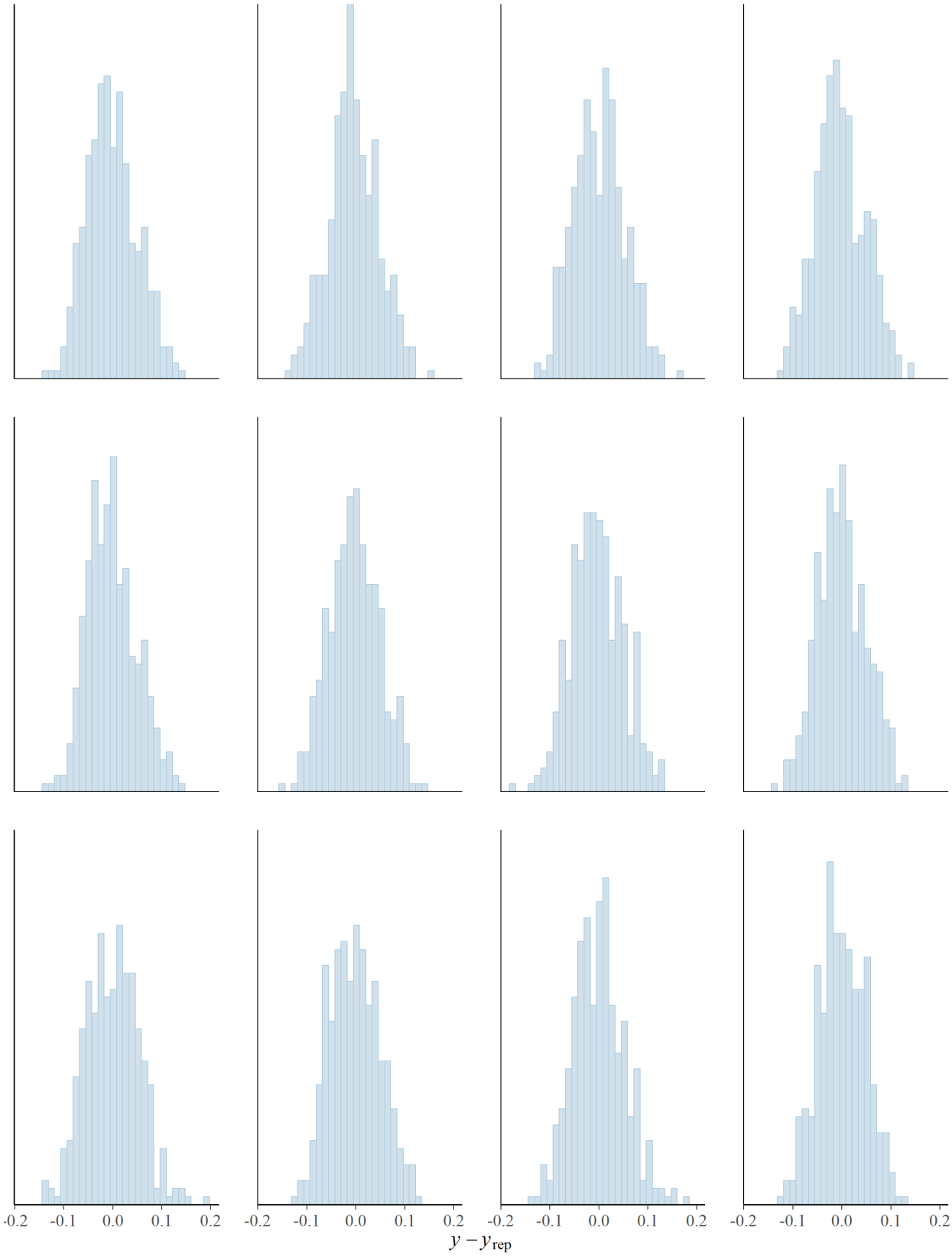


Figure 19: plot of chunk unnamed-chunk-9

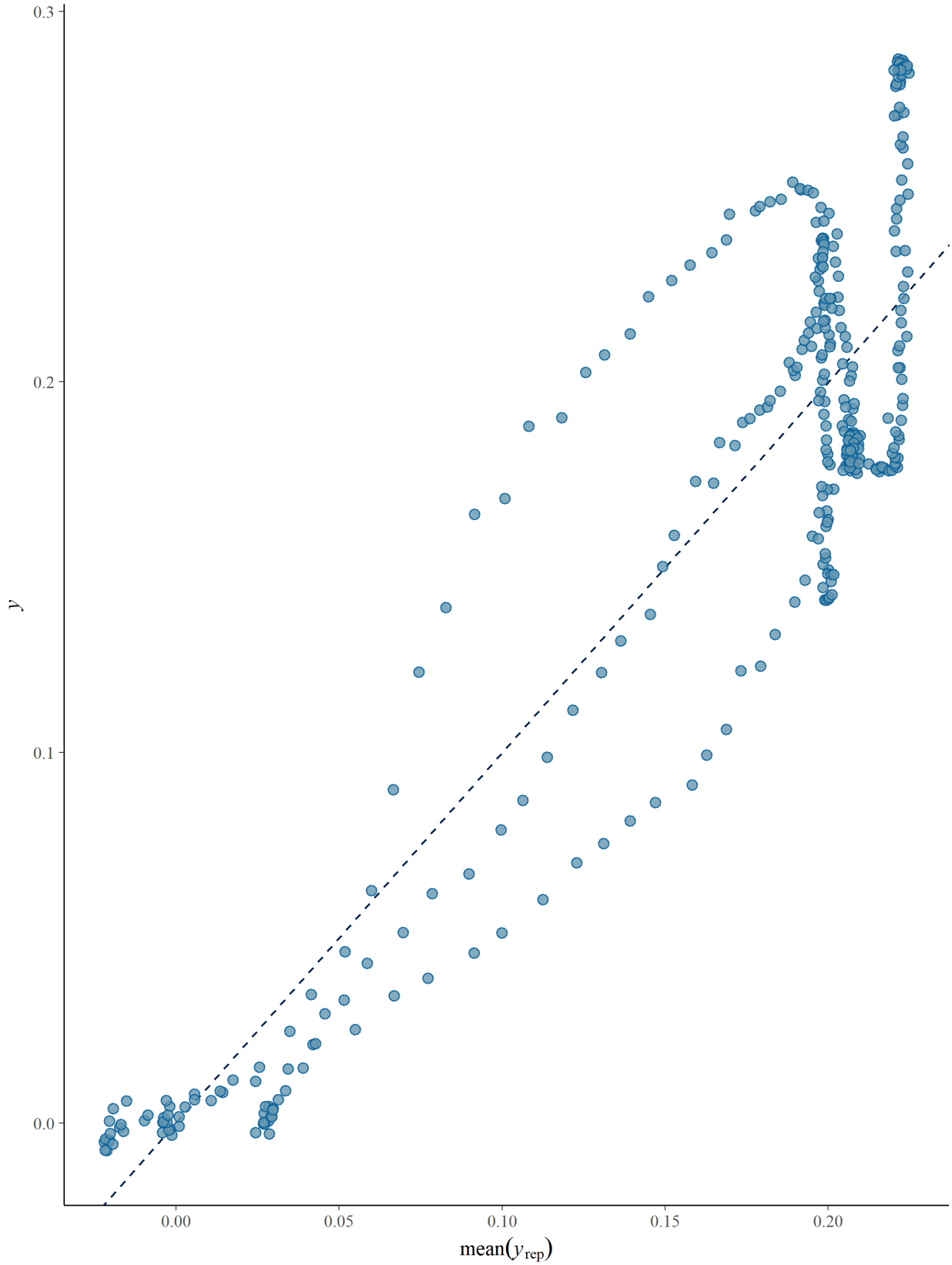


Figure 20: plot of chunk unnamed-chunk-9

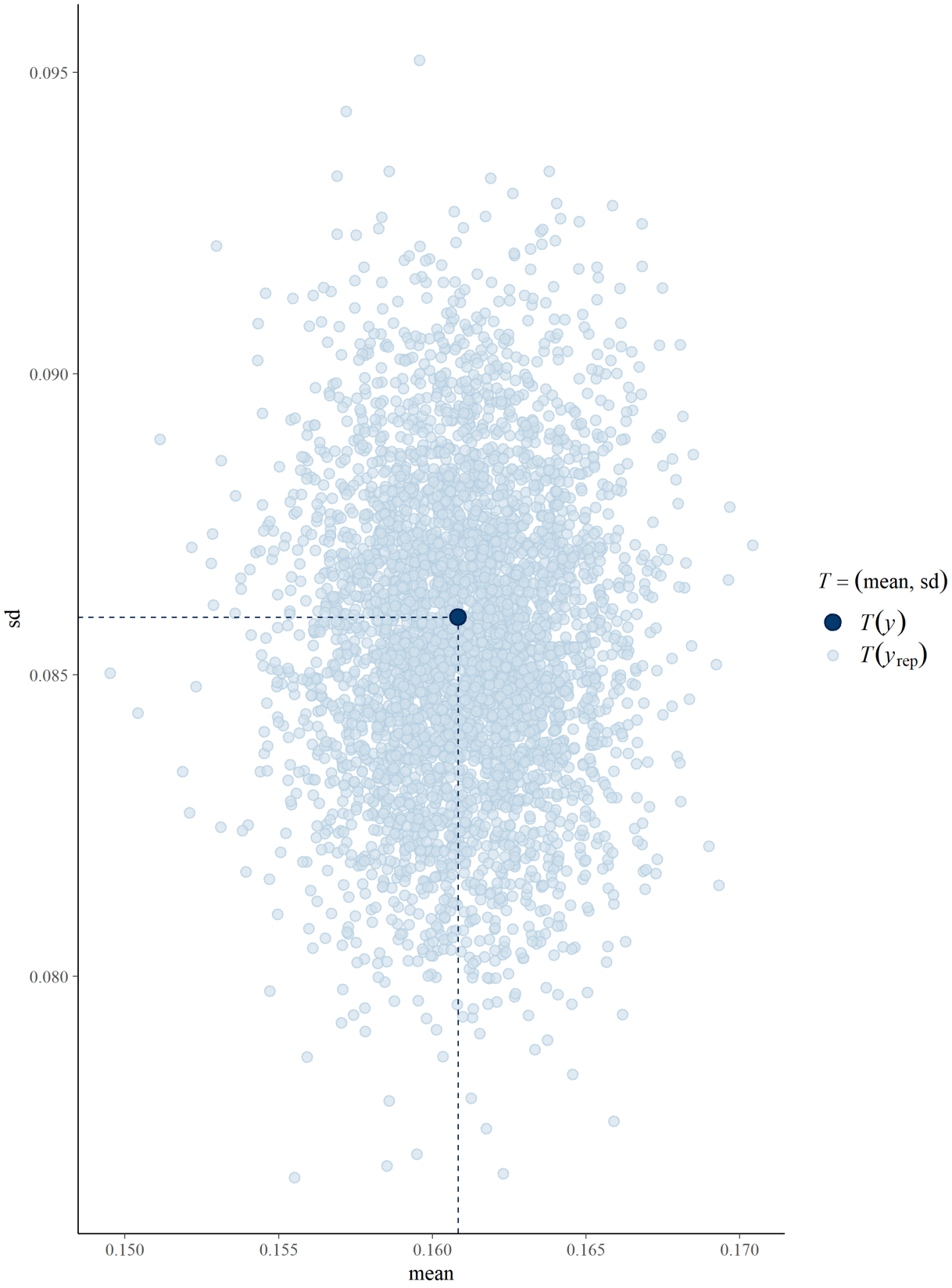


Figure 21: plot of chunk unnamed-chunk-9

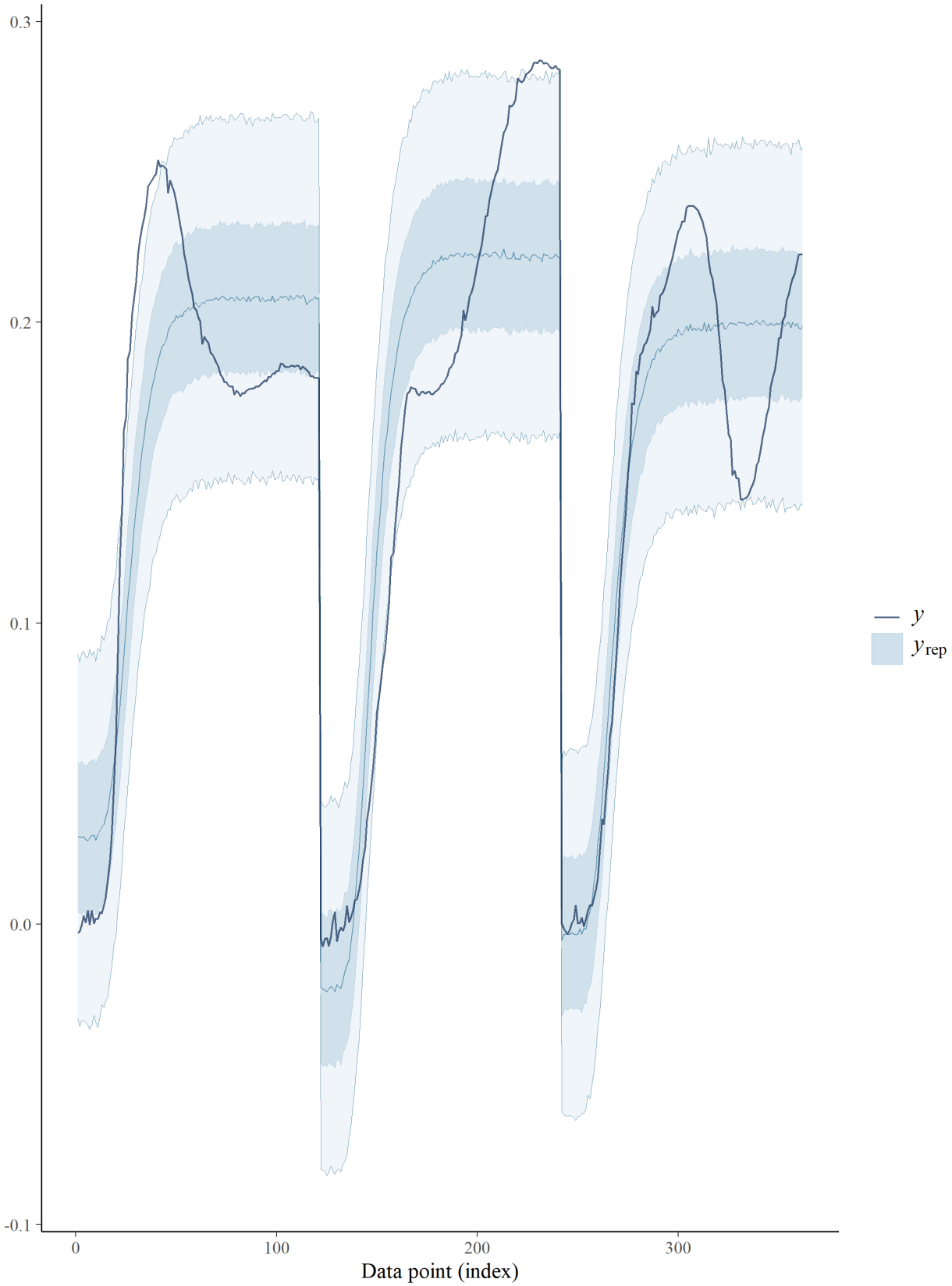


Figure 22: plot of chunk unnamed-chunk-9

```
plot(gsw~tmin, dd_init, pch=16, main = unique(dd_init$TW))
points(fitted(fit_init_mc)[,1]~tmin, dd_init, pch=16, col="red")
```

```
##look at the effects from the model
fixef(fit_init_mc)
```

```
                  Estimate  Est.Error        Q2.5      Q97.5
g0_Intercept     0.00625965 0.01058791 -0.01328097 0.02813052
g1_Intercept     0.21035988 0.01689921  0.17599889 0.24583332
k_Intercept      3.88407284 0.38295017  3.16617548 4.66817659
lambda_Intercept 8.24568233 0.59494921  7.02663964 9.38146679
```

```
ranef(fit_init_mc)
```

```
$Rep
, , g0_Intercept

      Estimate  Est.Error          Q2.5        Q97.5
1  0.020826705 0.01118073 -0.0007933152  0.042412796
2 -0.026582261 0.01205160 -0.0516963844 -0.005388827
3 -0.009744088 0.01140882 -0.0337264427  0.010767266

, , g1_Intercept

      Estimate  Est.Error        Q2.5      Q97.5
1 -0.002626639 0.01700963 -0.03784748 0.03258209
2  0.012315570 0.01707543 -0.02223718 0.04838185
3 -0.011298037 0.01699742 -0.04713921 0.02255815
```

```
coef(fit_init_mc)
```

```
$Rep
, , g0_Intercept

      Estimate   Est.Error         Q2.5        Q97.5
1  0.027086355 0.008517653  0.009999541  0.043644282
2 -0.020322611 0.008452965 -0.037267672 -0.004181686
3 -0.003484438 0.007777907 -0.019252925  0.011215327

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2077332 0.003954648 0.2000199 0.2155870
2 0.2226754 0.004485620 0.2137802 0.2314103
3 0.1990618 0.003947456 0.1914377 0.2067731

, , k_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 3.884073 0.3829502 3.166175 4.668177
2 3.884073 0.3829502 3.166175 4.668177
3 3.884073 0.3829502 3.166175 4.668177

, , lambda_Intercept

  Estimate Est.Error    Q2.5    Q97.5
```

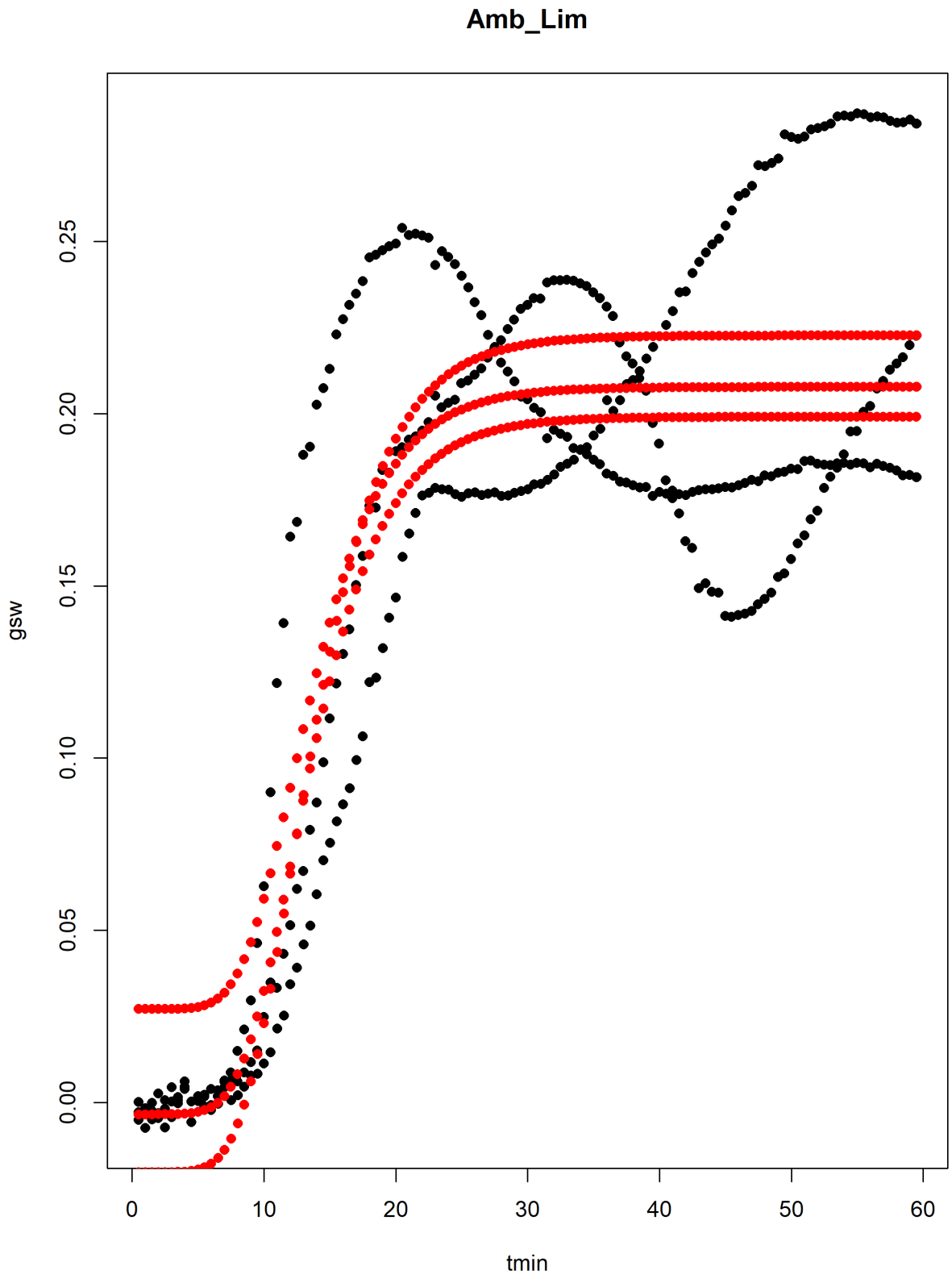


Figure 23: plot of chunk unnamed-chunk-9

```
1 8.245682 0.5949492 7.02664 9.381467
2 8.245682 0.5949492 7.02664 9.381467
3 8.245682 0.5949492 7.02664 9.381467
```

```
process <- function(d1, fit_template) {
  dd <- subset(d1, Light == "On" & th != 0 & !is.na(gsw))

  # t=0 at the light step
  dd$th <- dd$th - dd$th[1]
  dd$tmin <- dd$tmin - dd$tmin[1]

  ##remove the first time point, as seems to be impacted by change of lighting still
  dd <- subset(dd, th != 0)

  fit <- update(
    fit_template,
    newdata = dd,
    recompile = NULL,
    refresh = 0
  )

  # Check diagnostics, Important!
  print(summary(fit))
  rstan::check_hmc_diagnostics(fit$fit)
  #plot(fit)
  #pp_check(fit)
  # plot(gsw~th, dd, pch=16, main = unique(dd$TW))
  # points(fitted(fit)[,1]~th, dd, pch=16, col="red")

  return(fit)
}
```

Write a function to process each individual treatment

```
# Main loop that select each plant per species
fits <- list()

for (i in unique(li6800.dat$TW)) {
  d1 <- subset(li6800.dat, TW == i)

  cat("TW:", i, "\n")

  fits[[i]] <- process(d1, fit_init)
}
```

Implement loop

```
TW: Amb_Lim
```

```
Start sampling
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g1 + (g0 - g1) * exp(-((th/tau)^theta))
```

```
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         tau ~ 1
         theta ~ 1
   Data: dd (Number of observations: 362)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.01     0.01     0.03 1.00     2272     2227
sd(g1_Intercept)     0.02      0.02     0.01     0.07 1.00     1335     2141

Regression Coefficients:
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept       0.00      0.01    -0.02     0.03 1.00     1964     2416
g1_Intercept       0.21      0.02     0.18     0.25 1.01     1294     1387
tau_Intercept      0.26      0.01     0.24     0.27 1.00     3861     2643
theta_Intercept    3.63      0.39     2.94     4.45 1.00     3985     2714

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.03     0.04 1.00     3958     2935

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.

TW: Amb_Unlim

Start sampling

 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g1 + (g0 - g1) * exp(-((th/tau)^theta))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         tau ~ 1
         theta ~ 1
   Data: dd (Number of observations: 363)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000
```

```
Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.03      0.00     0.02     0.04 1.00     2654     2659
sd(g1_Intercept)     0.05      0.02     0.02     0.10 1.00     2164     2162

Regression Coefficients:
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept       0.01      0.01    -0.02     0.04 1.00     2793     2750
g1_Intercept       0.34      0.03     0.28     0.39 1.00     1864     2084
tau_Intercept      0.22      0.01     0.21     0.23 1.00     5055     2494
theta_Intercept    1.70      0.19     1.36     2.11 1.00     3993     3075

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.03      0.00     0.03     0.03 1.00     4593     2673

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.

TW: Heat_Lim

Start sampling

 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g1 + (g0 - g1) * exp(-((th/tau)^theta))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         tau ~ 1
         theta ~ 1
   Data: dd (Number of observations: 362)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.03      0.01     0.02     0.04 1.00     3029     2760
sd(g1_Intercept)     0.05      0.02     0.02     0.10 1.00     1681     2370

Regression Coefficients:
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

```
g0_Intercept        0.02      0.01    -0.01     0.04 1.00     2242     2207
g1_Intercept        0.33      0.03     0.27     0.39 1.00     1617     2094
tau_Intercept       0.12      0.00     0.12     0.13 1.00     3824     2900
theta_Intercept     3.28      0.36     2.66     4.07 1.00     3220     2230

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.04     0.04 1.00     4317     2674

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.

TW: Heat_Unlim

Start sampling

 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g1 + (g0 - g1) * exp(-((th/tau)^theta))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         tau ~ 1
         theta ~ 1
   Data: dd (Number of observations: 363)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.03      0.01     0.02     0.04 1.00     2663     2385
sd(g1_Intercept)     0.05      0.02     0.02     0.10 1.00     2374     2540

Regression Coefficients:
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept      -0.00      0.01    -0.03     0.03 1.00     2431     2415
g1_Intercept       0.45      0.03     0.39     0.51 1.00     1616     1533
tau_Intercept      0.30      0.01     0.28     0.32 1.00     3722     2375
theta_Intercept    0.99      0.04     0.91     1.07 1.00     3644     2753

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.02      0.00     0.02     0.02 1.00     4067     2290
```

```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.
```

```
pp_check(fits[[1]])  # compares density of the observed data with data generated from posterior predict
```

Explore model output

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fits[[2]])  # compares density of the observed data with data generated from posterior predict
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fits[[3]])  # compares density of the observed data with data generated from posterior predict
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fits[[4]])  # compares density of the observed data with data generated from posterior predict
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
##look at the effects from the model
coef(fits[[1]])
```

```
$Rep
, , g0_Intercept

      Estimate   Est.Error         Q2.5        Q97.5
1  0.024660967 0.008995836  0.007175031  0.041941528
2 -0.021198868 0.008574905 -0.038298706 -0.004643570
3 -0.005898066 0.007885002 -0.021206760  0.009111512

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2064017 0.003845503 0.1989923 0.2139802
2 0.2201927 0.004202515 0.2120440 0.2285581
3 0.1976826 0.003956981 0.1896084 0.2055666

, , tau_Intercept

   Estimate   Est.Error      Q2.5    Q97.5
1 0.2559939 0.006342612 0.2437064 0.268405
```

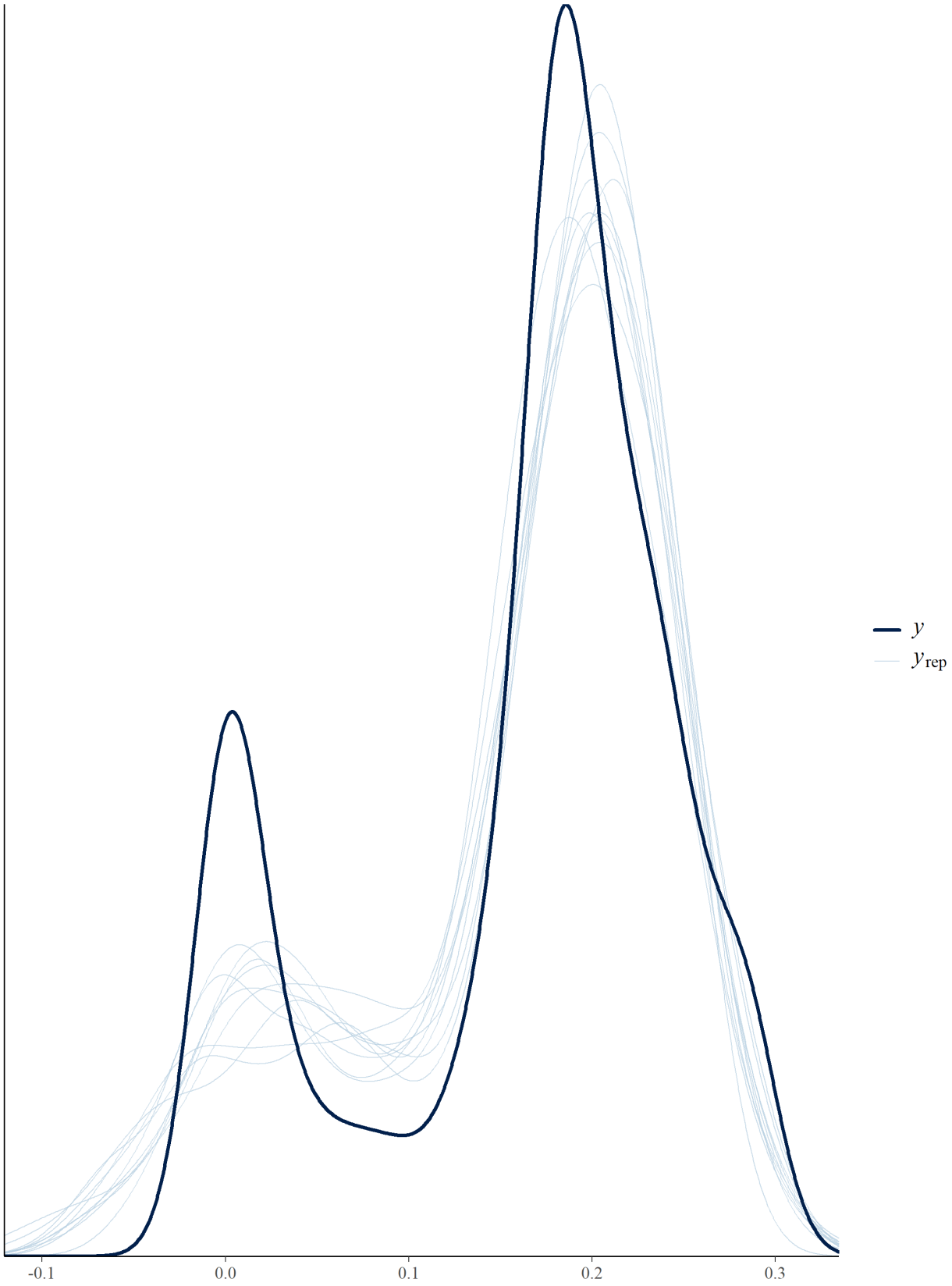


Figure 24: plot of chunk unnamed-chunk-12

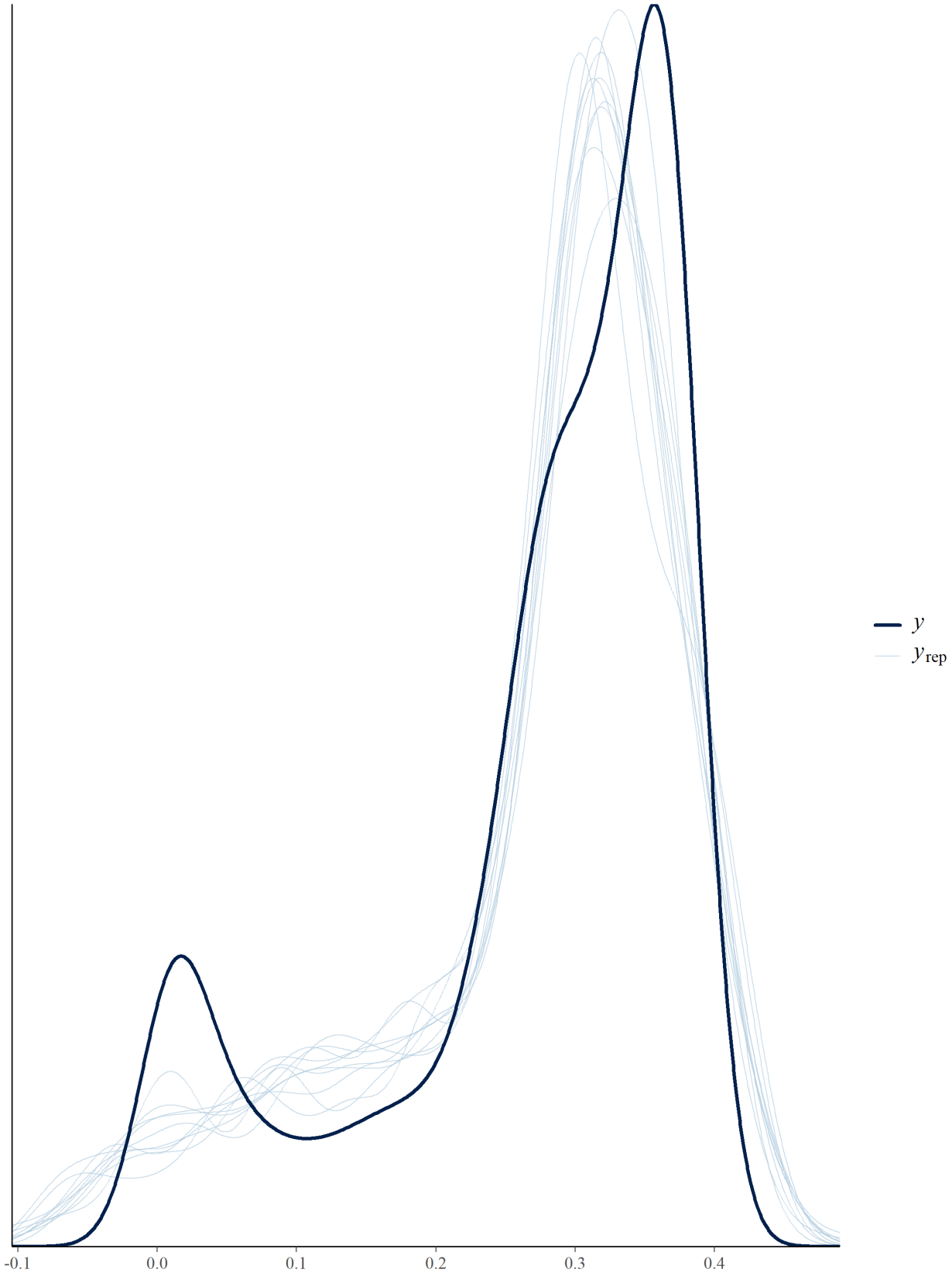


Figure 25: plot of chunk unnamed-chunk-12

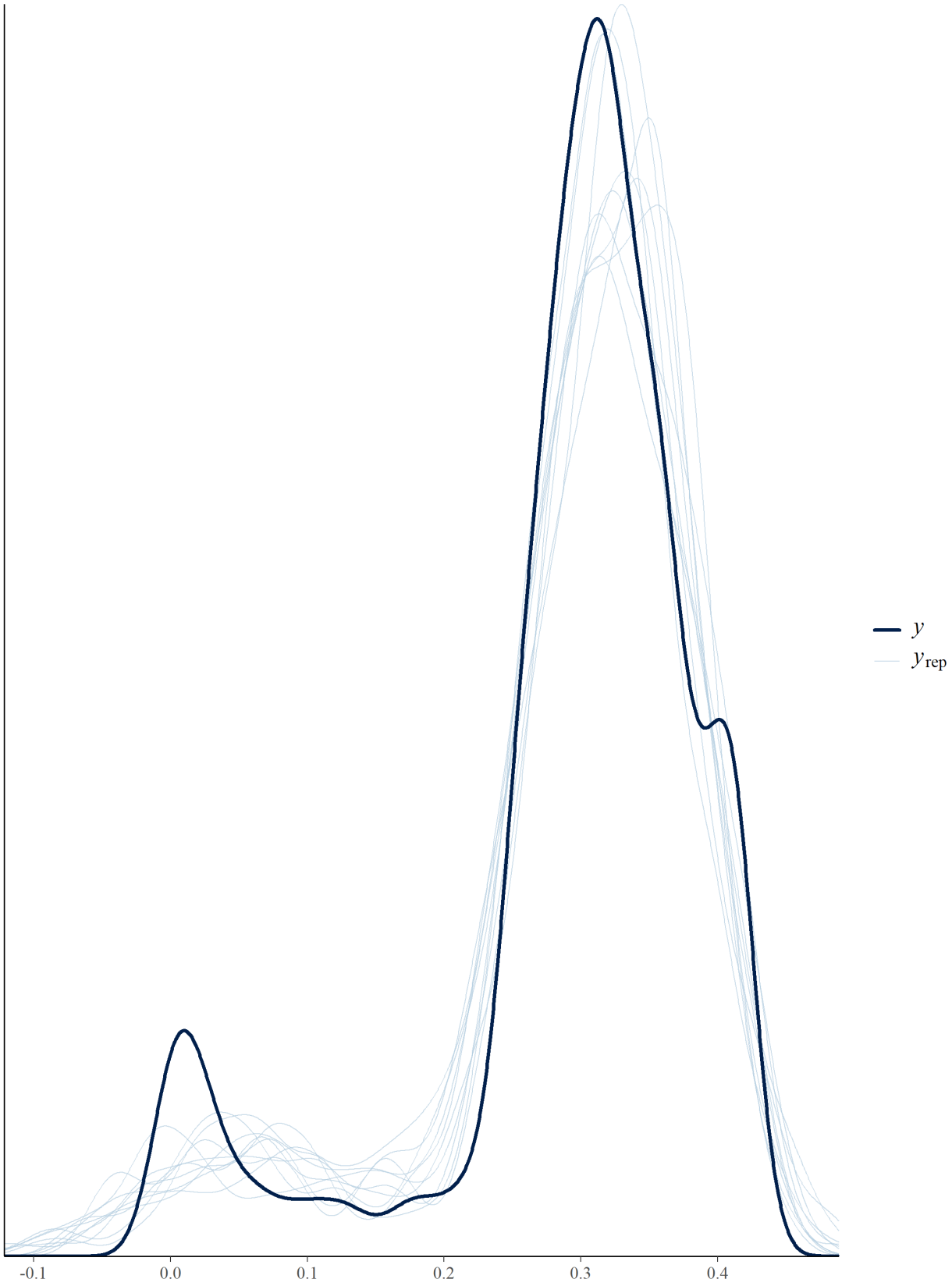


Figure 26: plot of chunk unnamed-chunk-12

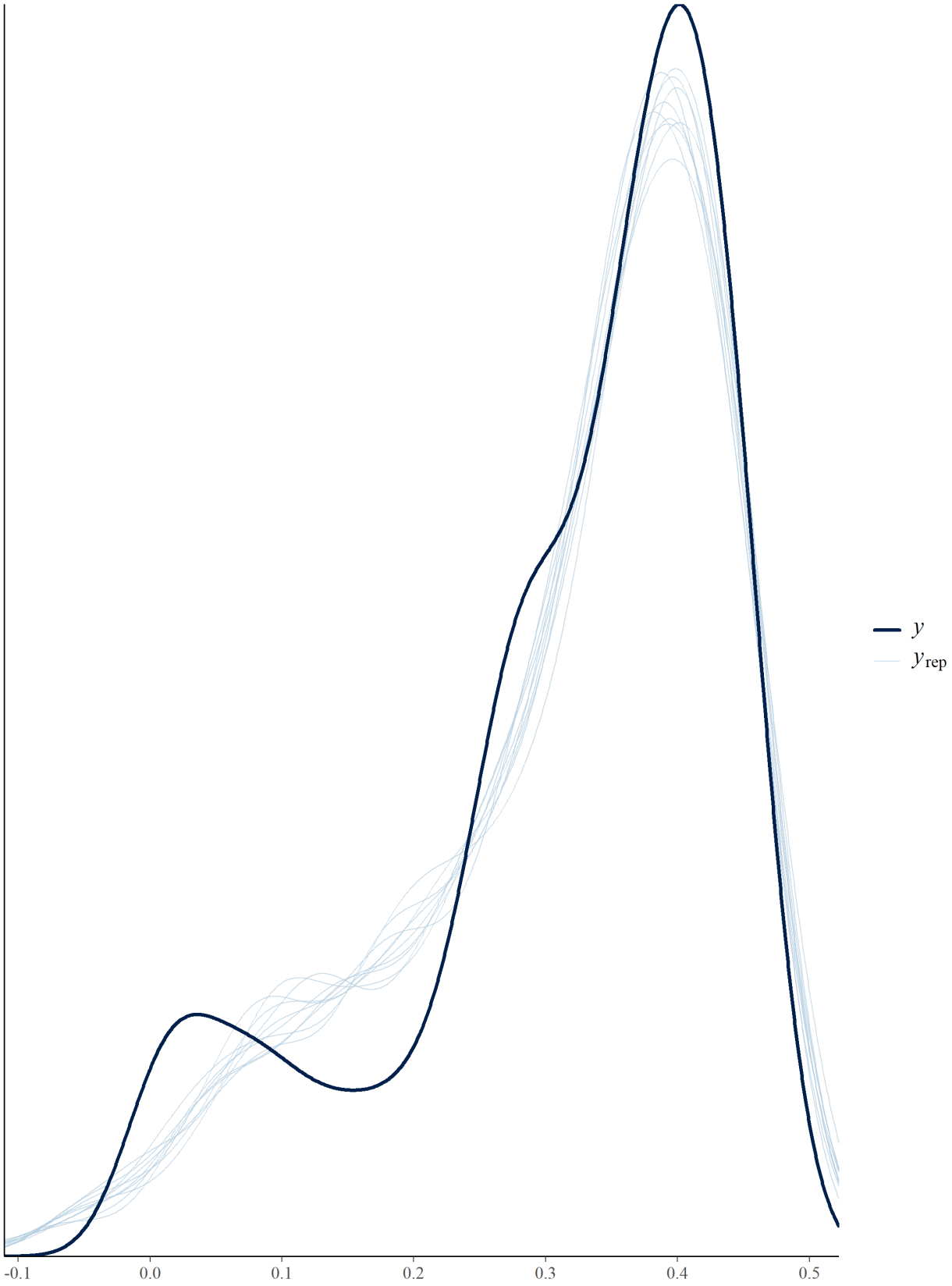


Figure 27: plot of chunk unnamed-chunk-12

```
2 0.2559939 0.006342612 0.2437064 0.268405
3 0.2559939 0.006342612 0.2437064 0.268405

, , theta_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 3.632883 0.3886937 2.935102 4.452607
2 3.632883 0.3886937 2.935102 4.452607
3 3.632883 0.3886937 2.935102 4.452607
```

```
coef(fits[[2]])
```

```
$Rep
, , g0_Intercept

     Estimate  Est.Error        Q2.5       Q97.5
1 -0.03860993 0.01085082 -0.06022301 -0.01800608
2 -0.03424373 0.01212050 -0.06003578 -0.01118834
3  0.07383995 0.01122548  0.05185288  0.09584318

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.3229637 0.004323453 0.3148338 0.3318390
2 0.3123184 0.004701688 0.3035390 0.3219019
3 0.3804150 0.004158233 0.3723897 0.3888230

, , tau_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2210942 0.005498384 0.2108463 0.2317974
2 0.2210942 0.005498384 0.2108463 0.2317974
3 0.2210942 0.005498384 0.2108463 0.2317974

, , theta_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 1.697553 0.1941195 1.362771 2.110924
2 1.697553 0.1941195 1.362771 2.110924
3 1.697553 0.1941195 1.362771 2.110924
```

```
coef(fits[[3]])
```

```
$Rep
, , g0_Intercept

     Estimate  Est.Error         Q2.5        Q97.5
1  0.01938510 0.01319793 -0.007329645  0.044903460
2  0.05413860 0.01397053  0.027455813  0.081056822
3 -0.03677645 0.01411761 -0.064051347 -0.008600166

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2920438 0.003802210 0.2844037 0.2994957
2 0.3663368 0.003863137 0.3587615 0.3739955
```

```
3 0.3231305 0.003816369 0.3155947 0.3304594

, , tau_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.1233718 0.003128216 0.1173038 0.1295427
2 0.1233718 0.003128216 0.1173038 0.1295427
3 0.1233718 0.003128216 0.1173038 0.1295427

, , theta_Intercept

  Estimate Est.Error    Q2.5    Q97.5
1 3.284134 0.3563351 2.65882 4.068297
2 3.284134 0.3563351 2.65882 4.068297
3 3.284134 0.3563351 2.65882 4.068297
```

```r
coef(fits[[4]])
```

```
$Rep
, , g0_Intercept

       Estimate   Est.Error         Q2.5       Q97.5
1  0.0004299699 0.009476602 -0.018907696  0.01884583
2  0.0253235111 0.008633175  0.008074516  0.04209427
3 -0.0936482990 0.010623954 -0.114726921 -0.07350714

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.4773796 0.007135766 0.4642401 0.4928296
2 0.4095112 0.006240908 0.3979430 0.4225171
3 0.4653220 0.008500533 0.4501630 0.4840035

, , tau_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2999061 0.009427406 0.2828145 0.3193698
2 0.2999061 0.009427406 0.2828145 0.3193698
3 0.2999061 0.009427406 0.2828145 0.3193698

, , theta_Intercept

   Estimate  Est.Error      Q2.5   Q97.5
1 0.9918428 0.04208853 0.9122033 1.07417
2 0.9918428 0.04208853 0.9122033 1.07417
3 0.9918428 0.04208853 0.9122033 1.07417
```

```r
# Main loop that select each plant per species
fits_mc <- list()

for (i in unique(li6800.dat$TW)) {
  d1 <- subset(li6800.dat, TW == i)

  cat("TW:", i, "\n")
```

```
  fits_mc[[i]] <- process(d1, fit_init_mc)
}
```

Implement loop - McAusland

```
TW: Amb_Lim

Start sampling

 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         k ~ 1
         lambda ~ 1
   Data: dd (Number of observations: 362)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.01     0.01     0.03 1.00     1983     1784
sd(g1_Intercept)     0.02      0.02     0.01     0.07 1.00     1120      985

Regression Coefficients:
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept         0.01      0.01    -0.01     0.03 1.00     2101     2450
g1_Intercept         0.21      0.02     0.18     0.25 1.00     1275      838
k_Intercept          3.89      0.41     3.13     4.76 1.00     2701     2023
lambda_Intercept     8.24      0.64     6.91     9.44 1.00     2549     1731

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.03     0.04 1.00     3571     2639

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.

TW: Amb_Unlim

Start sampling
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         k ~ 1
         lambda ~ 1
   Data: dd (Number of observations: 363)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.03      0.00     0.02     0.04 1.00     2682     2767
sd(g1_Intercept)     0.05      0.02     0.02     0.10 1.00     1898     2328

Regression Coefficients:
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept         0.02      0.01    -0.01     0.04 1.00     2384     2368
g1_Intercept         0.34      0.03     0.28     0.40 1.00     1485     1488
k_Intercept          5.57      0.38     4.83     6.34 1.00     2528     2189
lambda_Intercept     3.37      0.70     1.92     4.66 1.00     2473     1750

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.03      0.00     0.03     0.03 1.00     4180     2637

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.

TW: Heat_Lim

Start sampling

 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         k ~ 1
         lambda ~ 1
   Data: dd (Number of observations: 362)
```

```
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.01     0.02     0.04 1.00     2480     2394
sd(g1_Intercept)     0.05      0.02     0.02     0.10 1.00     1960     2584

Regression Coefficients:
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept         0.02      0.01    -0.00     0.05 1.00     2218     2435
g1_Intercept         0.33      0.03     0.26     0.39 1.00     1437     1844
k_Intercept          1.82      0.18     1.47     2.17 1.00     1973     2302
lambda_Intercept     4.02      0.34     3.34     4.66 1.00     2154     1754

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.04      0.00     0.04     0.04 1.00     3553     2801

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.

TW: Heat_Unlim

Start sampling

 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + (1 | Rep)
         g1 ~ 1 + (1 | Rep)
         k ~ 1
         lambda ~ 1
   Data: dd (Number of observations: 363)
  Draws: 4 chains, each with iter = 3000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.03      0.00     0.02     0.04 1.00     3119     2578
sd(g1_Intercept)     0.04      0.02     0.02     0.09 1.00     2180     2046
```

```
Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
g0_Intercept         0.01      0.01    -0.02     0.04 1.00     2381     2336
g1_Intercept         0.41      0.03     0.35     0.47 1.00     1848     1901
k_Intercept          9.12      0.23     8.67     9.59 1.00     4481     2671
lambda_Intercept     0.18      0.17     0.01     0.62 1.00     2830     1895

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.03      0.00     0.02     0.03 1.00     4743     2834

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

Divergences:

0 of 4000 iterations ended with a divergence.


Tree depth:

0 of 4000 iterations saturated the maximum tree depth of 12.


Energy:

E-BFMI indicated no pathological behavior.
```

```
pp_check(fits_mc[[1]])  # compares density of the observed data with data generated from posterior pred
```

Explore model output

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fits_mc[[2]])  # compares density of the observed data with data generated from posterior pred
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fits_mc[[3]])  # compares density of the observed data with data generated from posterior pred
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fits_mc[[4]])  # compares density of the observed data with data generated from posterior pred
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
##look at the effects from the model
coef(fits_mc[[1]])
```

```
$Rep
, , g0_Intercept

      Estimate   Est.Error        Q2.5        Q97.5
1  0.027154330 0.008575412  0.01031932  0.044055490
2 -0.020259833 0.008410331 -0.03695759 -0.004404156
3 -0.003602326 0.007640791 -0.01899226  0.011395916
```

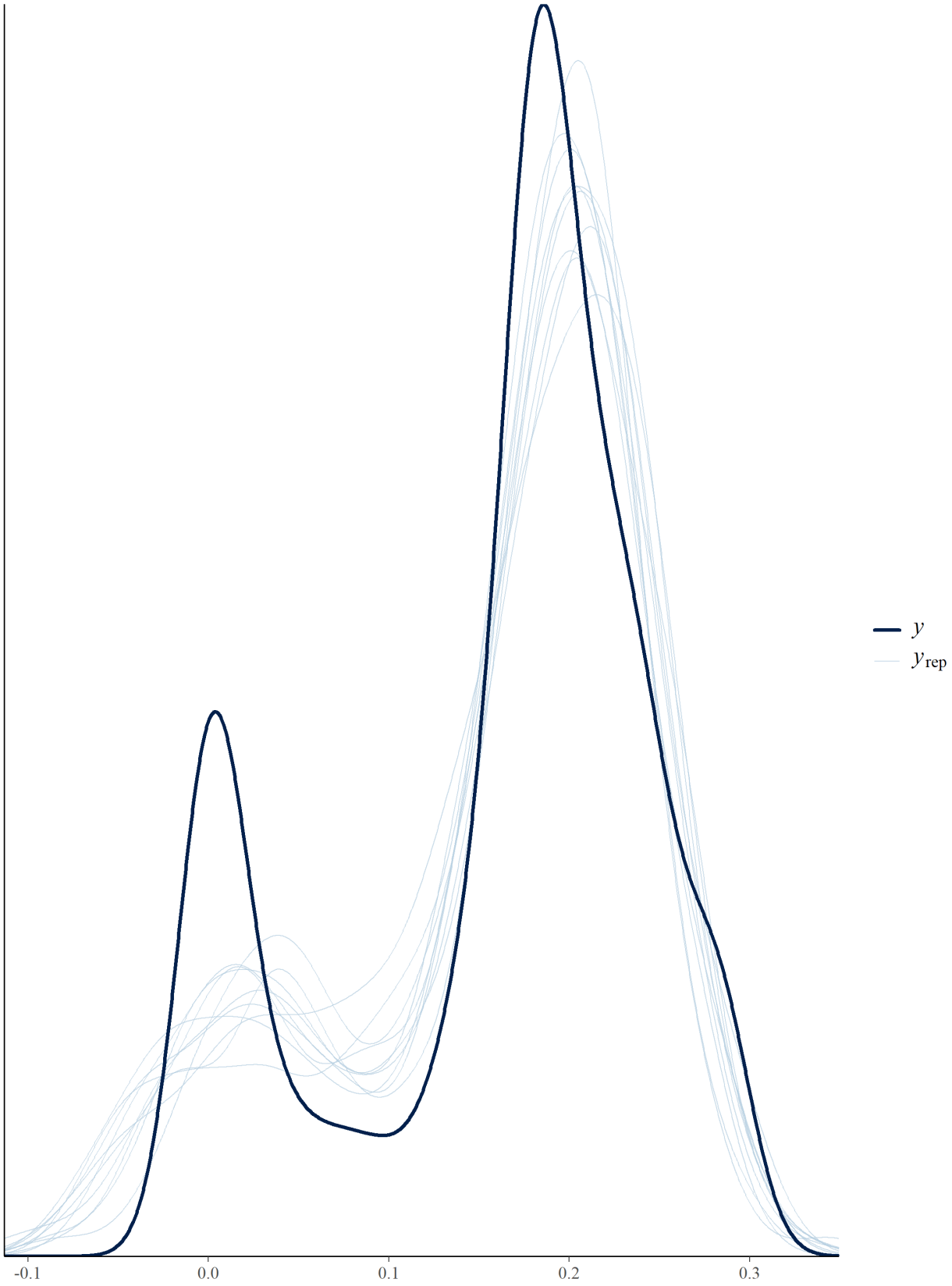


Figure 28: plot of chunk unnamed-chunk-14

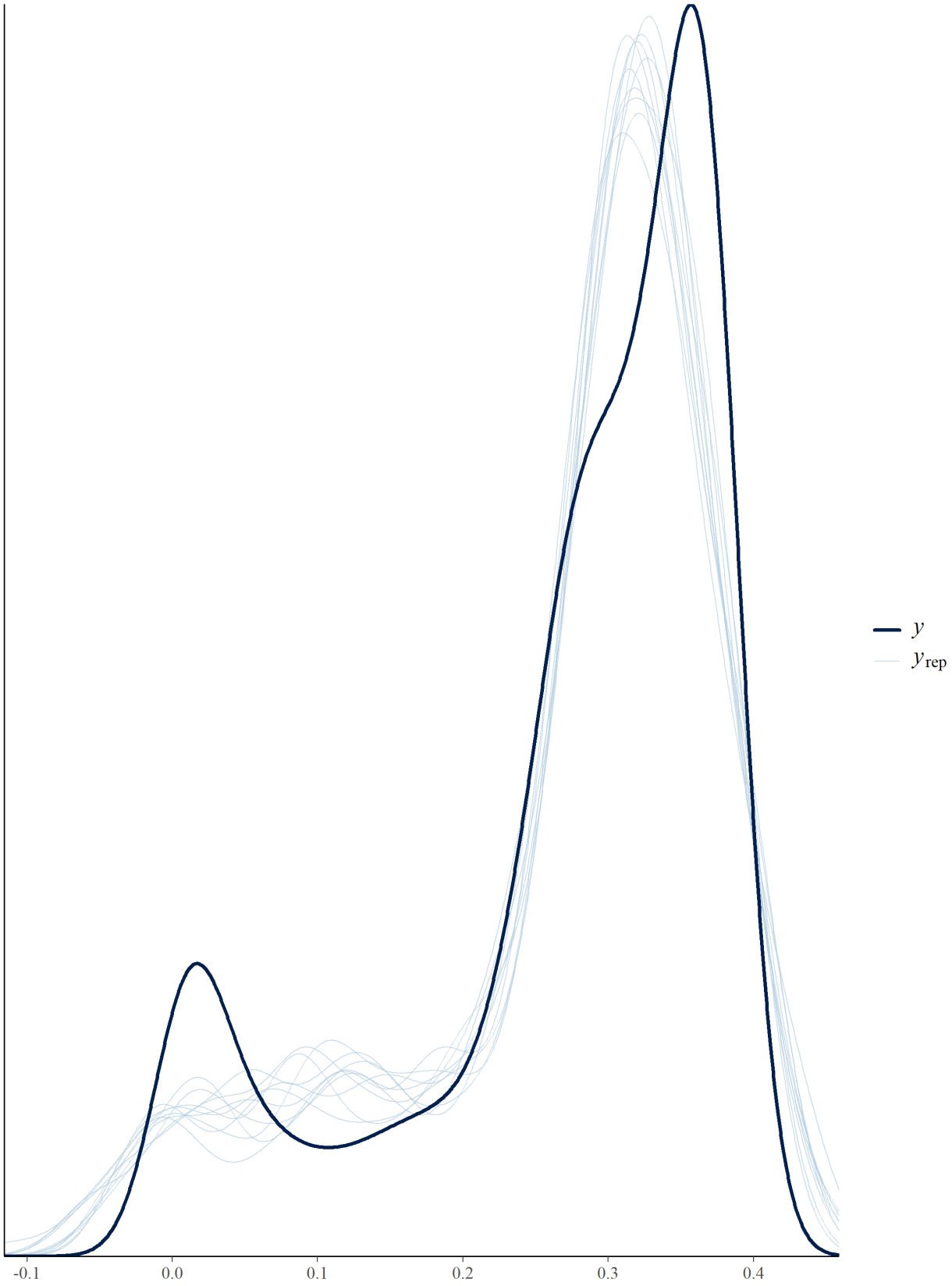


Figure 29: plot of chunk unnamed-chunk-14

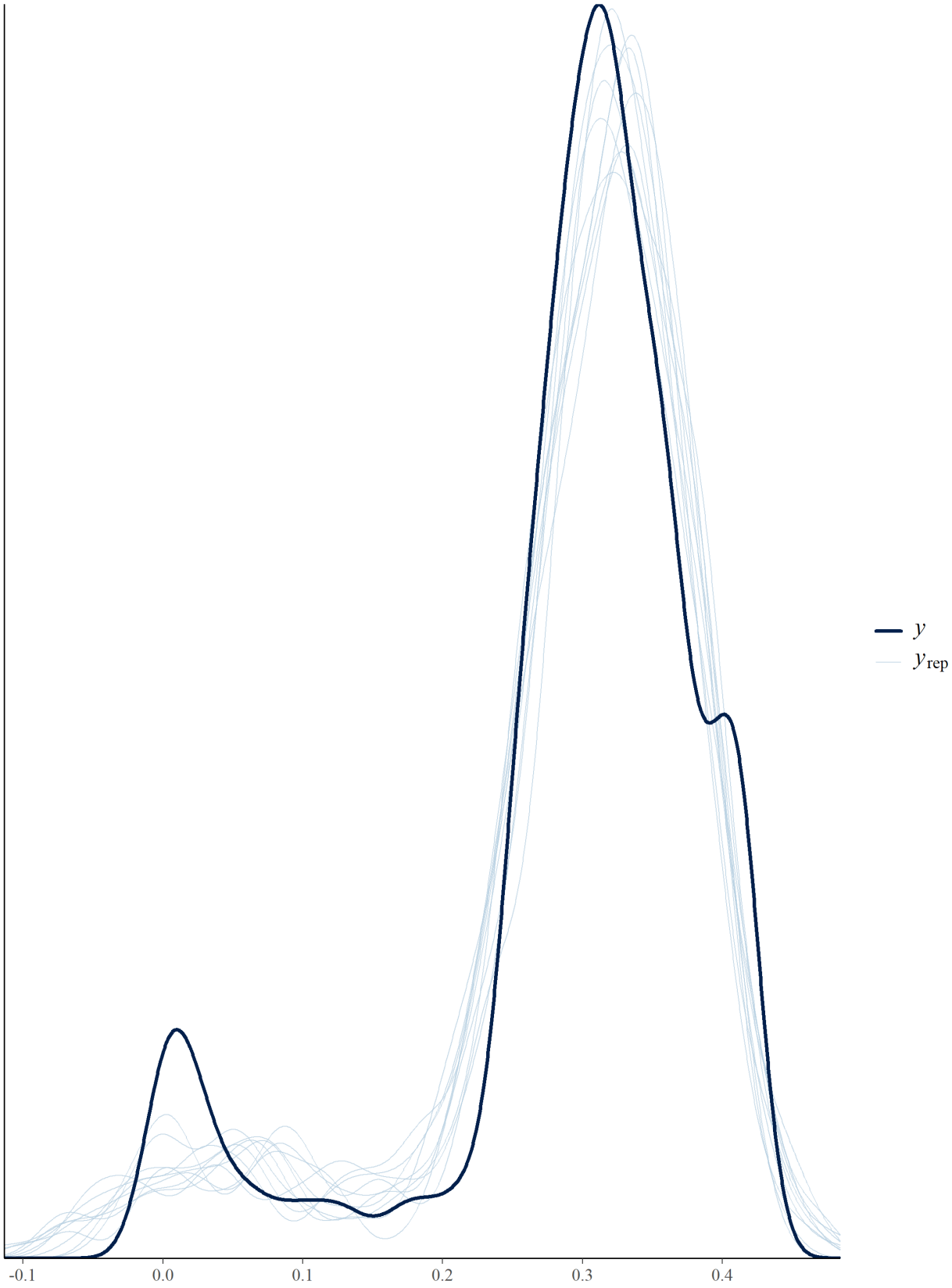


Figure 30: plot of chunk unnamed-chunk-14

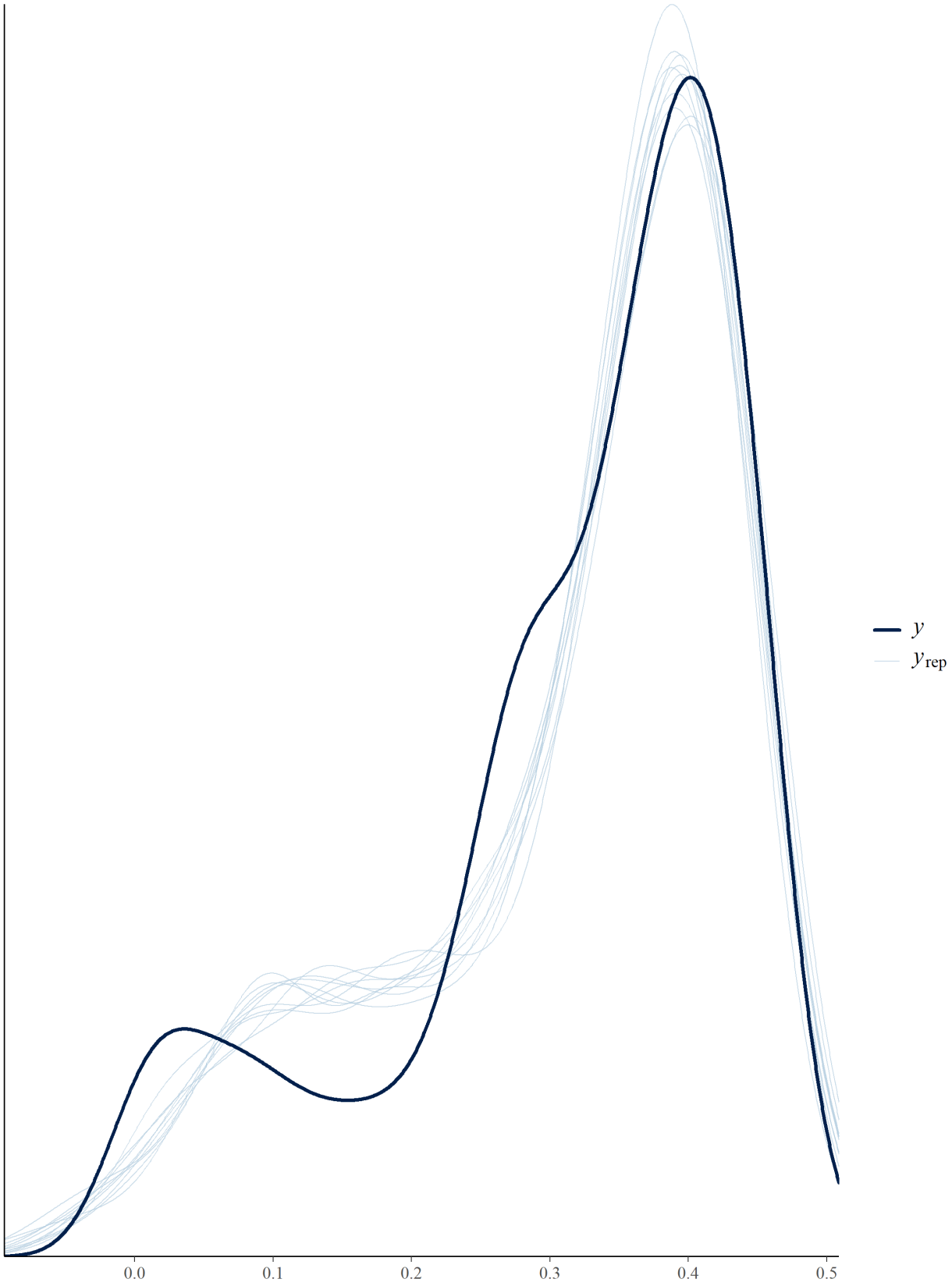


Figure 31: plot of chunk unnamed-chunk-14

```
, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2077605 0.003841499 0.2000201 0.2154027
2 0.2226781 0.004401473 0.2141500 0.2312090
3 0.1991932 0.004026528 0.1912127 0.2070279

, , k_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 3.890118  0.405581 3.129246 4.757741
2 3.890118  0.405581 3.129246 4.757741
3 3.890118  0.405581 3.129246 4.757741

, , lambda_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 8.240774 0.6368175 6.908571 9.435375
2 8.240774 0.6368175 6.908571 9.435375
3 8.240774 0.6368175 6.908571 9.435375
```

```
coef(fits_mc[[2]])
```

```
$Rep
, , g0_Intercept

     Estimate  Est.Error        Q2.5        Q97.5
1 -0.03040110 0.01107837 -0.05331100 -0.009935962
2 -0.02447592 0.01116858 -0.04758602 -0.004229508
3  0.08358140 0.01096312  0.06056172  0.103491024

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.3229345 0.003427281 0.3160712 0.3294829
2 0.3120514 0.003605522 0.3049013 0.3189584
3 0.3798730 0.003402423 0.3732285 0.3865912

, , k_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 5.570316 0.3843222 4.833788 6.341788
2 5.570316 0.3843222 4.833788 6.341788
3 5.570316 0.3843222 4.833788 6.341788

, , lambda_Intercept

  Estimate Est.Error     Q2.5   Q97.5
1 3.372691 0.6997213 1.917471 4.65887
2 3.372691 0.6997213 1.917471 4.65887
3 3.372691 0.6997213 1.917471 4.65887
```

```
coef(fits_mc[[3]])
```

```
$Rep
```

```
, , g0_Intercept

     Estimate  Est.Error        Q2.5       Q97.5
1  0.02684227 0.01238221  0.00228397  0.05146800
2  0.06210048 0.01426848  0.03373945  0.08978361
3 -0.02901219 0.01350788 -0.05615944 -0.00303010

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.2925288 0.003837467 0.2850951 0.3000701
2 0.3666841 0.003916438 0.3590314 0.3741826
3 0.3237615 0.003981377 0.3157955 0.3318069

, , k_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 1.815592  0.179213 1.468473 2.166501
2 1.815592  0.179213 1.468473 2.166501
3 1.815592  0.179213 1.468473 2.166501

, , lambda_Intercept

  Estimate Est.Error     Q2.5    Q97.5
1 4.022777 0.3352667 3.342334 4.664478
2 4.022777 0.3352667 3.342334 4.664478
3 4.022777 0.3352667 3.342334 4.664478
```

```
coef(fits_mc[[4]])
```

```
$Rep
, , g0_Intercept

     Estimate   Est.Error         Q2.5       Q97.5
1  0.01919789 0.008215154  0.002993877  0.03577236
2  0.04360957 0.007496140  0.029090915  0.05825063
3 -0.06683475 0.008451857 -0.083561969 -0.05022606

, , g1_Intercept

   Estimate   Est.Error      Q2.5     Q97.5
1 0.4431925 0.003634746 0.4360435 0.4503558
2 0.3809679 0.003569734 0.3740937 0.3881208
3 0.4240267 0.004125460 0.4160401 0.4322197

, , k_Intercept

  Estimate Est.Error     Q2.5   Q97.5
1 9.119739 0.2324035 8.666456 9.58771
2 9.119739 0.2324035 8.666456 9.58771
3 9.119739 0.2324035 8.666456 9.58771

, , lambda_Intercept

   Estimate Est.Error        Q2.5     Q97.5
```

```
1 0.1820738 0.1679105 0.005453178 0.6223948
2 0.1820738 0.1679105 0.005453178 0.6223948
3 0.1820738 0.1679105 0.005453178 0.6223948
```

```
##calculate Slmax
Slmax.calc <- function(coef){
  ##pull out ceofficients, as they relate to parameters in Woning et al 2026
  gi <- coef$Rep[, "Estimate", "g0_Intercept"]
  gf <- coef$Rep[, "Estimate", "g1_Intercept"]
  lambda_c <- coef$Rep[, "Estimate", "theta_Intercept"]
  tau_c <- coef$Rep[, "Estimate", "tau_Intercept"]

  numerator_calc <- -1*(gi - gf) * lambda_c * exp((lambda_c - 1)/lambda_c)
  denominator_calc <- ((lambda_c - 1)/lambda_c)^(1/lambda_c) * tau_c

  numerator_calc/denominator_calc
}

Slmax.calc(coef(fits[[1]]))
```

```
       1        2        3
5.817154 7.726455 6.516205
```

```
Slmax.calc(coef(fits[[2]]))
```

```
       1        2        3
7.070217 6.776683 5.994774
```

```
Slmax.calc(coef(fits[[3]]))
```

```
       1        2        3
16.25164 18.60836 21.45201
```

```
Slmax.calc(coef(fits[[4]]))
```

```
  1   2   3
NaN NaN NaN
```

```
##calculate Slmax
Slmax.calc.mc <- function(coef){
  ##pull out ceofficients, as they relate to parameters in McAusland 2016
  gi <- coef$Rep[, "Estimate", "g0_Intercept"]
  gf <- coef$Rep[, "Estimate", "g1_Intercept"]
  k <- coef$Rep[, "Estimate", "k_Intercept"]

  (gf - gi)*k/exp(1)
}

Slmax.calc.mc(coef(fits_mc[[1]]))
```

```
        1         2         3
0.2584645 0.3476672 0.2902196
```

```
Slmax.calc.mc(coef(fits_mc[[2]]))
```

```
        1         2         3
0.7240570 0.6896134 0.6071622
```

```
Slmax.calc.mc(coef(fits_mc[[3]]))
```

```
        1         2         3
0.1774571 0.2034372 0.2356242
```

```
Slmax.calc.mc(coef(fits_mc[[4]]))
```

```
       1        2        3
1.422487 1.131825 1.646823
```

Save data

```
save.image(saveFile)
```